Performance Limits of Shadowing Effects-Based Passive Target Localization Assisted by Active Sensor Calibration via Visible Light Positioning

Xu Bao, Jiawei Zhou and Licheng Zhang

***Abstract*—In this paper, we aim at exploring the performance limits of passive indoor target localization utilizing multiple light-emitting diode (LED) transmitters embedded into the ceiling and multiple photodiode (PD) sensors anchored on the floor, where the precise locations of the PDs are not predetermined. This work enriches our understanding of error evolution in the time-domain localization cooperation. We propose a novel solution to estimate the target position by identifying the PDs occluded by the target, through measuring the changes in received signal strength (RSS) caused by the shadowing effects induced by this indoor target. Concurrently, these LEDs are used for the positional calibration of the PDs involved in the localization process. Specifically, we derive the Cramér-Rao lower bound (CRLB) on the estimation errors for both target localization and sensor position calibration, and then conduct the convergence and asymptotic performance analyses based on the obtained error bounds. Finally, simulation results corroborate the performance analysis, elucidating how the localization error evolves until it converges, and identifying the factors that influence this intrinsic convergence mechanism. The derived CRLBs and the long-term localization performance from these bounds provide a theoretical basis for optimizing passive positioning algorithms without requiring any auxiliary receiver.**

***Index Terms*—Visible light positioning, device-free localization, performance limits, Cramér-Rao lower bound, error evolution.**

I. Introduction

VISIBLE light positioning (VLP) is regarded as a promising solution for indoor positioning scenarios, leveraging the widespread use of light-emitting diodes (LEDs) in interior illumination as reliable transmitters. These LEDs can offer low power consumption and low cost, ensuring VLP particularly advantageous in scenarios where the global positioning system (GPS) is obstructed or inaccessible [1]. Due to the high data rate and low cost of photodiodes (PDs) compared with image sensors, most current VLP solutions utilize PDs as receivers to capture and process signal information from LEDs. Among the specific methods for VLP, there is a preference for a two-step approach, which offers low computational complexity by extracting relevant parameters from the received signals before applying them to position estimation. A detailed survey of this VLP method is provided in [2].

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Received signal strength (RSS) is widely favored for positional estimation due to its cost-effectiveness and the absence of synchronization requirements. The reliability of the Lambertian formula in quantifying visible light channel attenuation further solidifies RSS as the dominant discriminant parameter in VLP [3]. Among various position estimation methods, statistical approaches are particularly effective, offering asymptotic optimization of VLP performance by fully leveraging the statistical properties of parameter measurements. Hence, the VLP approach based on statistical models and RSS measurements has become the prevailing solution for conducting the performance evaluation, as demonstrated in [4].

In contrast to active localization approaches, which require the target to carry a receiver to collect signals from different LEDs, RSS-based passive VLP eliminates the need for detection equipment on the target, depending instead on multiple sensors in the environment. The active VLP's requirement for target-mounted devices reduces its competitiveness in scenarios where convenience and privacy are paramount. Conversely, passive VLP enables device-free localization (DFL) without any receiver, offering a non-intrusive and privacy-preserving alternative [5]. This approach is particularly advantageous in applications such as personnel intrusion detection and remote elder care [6]. In this paper, we try to introduce a standardized performance analysis scheme for passive VLP based on RSS observations and statistical estimation, aiming to explore the fundamental limitations of the positioning performance.

In general, the localization performance of RSS-based VLP, a method that typically relies on a signal propagation model such as the Lambertian channel, is influenced by the measurement signals and the measurement noise [7]. Key parameters embedded in the measurement signals, which are associated with the target's position, play a critical role in the impact of measurement variations. Additionally, external factors, including the deployment coverage of LEDs and PDs, the movement characteristics of the target, and prior knowledge of the target's previous positions, can significantly affect VLP performance [8]. Therefore, it is crucial to uncover the performance limits of RSS-based passive VLP systems, as this understanding is essential for designing effective localization algorithms.

The Cramér-Rao lower bound (CRLB), which offers a lower bound on the variance of an unbiased estimator, is the currently prevailing performance benchmark for evaluating VLP error [9]. In this paper, we concentrate on the CRLB to reveal the performance mechanism of RSS-based passive VLP systems. We begin by reviewing recent advancements in passive VLP and VLP performance analysis to motivate our work, followed by a detailed discussion of the contributions in this paper.

A. Related Works

Despite numerous studies on passive VLP implementations over the past five years, the performance limits of RSS-based passive VLP—particularly its long-term performance and the influencing factors—remain not fully known, as shown below.

1) Passive VLP: In [10], PDs on the wall are used to detect changes in visible light attenuation caused by moving targets, enabling the identification of blocked PDs. However, the PDs need to be weighted before being used for precise positioning. [11] analyzes the performance bounds of a reflection-based passive VLP, but it relies on a mirror attached to the target to enhance light reflection, which may be somewhat idealistic. [12] employs wall-mounted PDs to detect changes in RSS of visible light, but still requires RSS fingerprinting for localization. Although the deep learning-based passive VLP proposed in [13] eliminates the need for knowledge of the visible light channel model, the data training process remains a significant challenge. Similarly, the fingerprinting method in [14] incurs high labor costs in maintaining database accuracy, making it less desirable. [15] offers a typical passive VLP scheme using the shaded links, where detecting RSS changes caused by the occlusion of ground-level PDs as a target enters the room is relatively feasible, though no performance bounds are given. Although [16] derives a passive VLP channel model based on the Lambertian pattern and RSS, it relies on the more ideal specular reflection characteristics of the received signal, which are still vulnerable to attenuation interference. A recent study of RSS-based passive VLP using shadowing effects, presented in [17], builds on the shaded links in [15] and addresses the previously missing VLP error bounds. However, [17] does not account for potential long-term performance variations due to target movement, a gap that our work aims to address.

2) CRLB Analysis: While limited work has been conducted to reveal the performance of passive VLP, numerous studies have explored the active VLP performance based on CRLBs, which can inspire the performance analysis of passive VLP. In [18], the CRLB for RSS-based VLP is examined under the assumption that the receiver comprises several PDs. The limits of quasi-synchronous VLP systems are analyzed in [19], even though the CRLB is applied to the direct localization method. [20] derives a CRLB for RSS-based simultaneous positioning and orientation (SPO) algorithms, where the PD orientation is constrained by the rotation matrix. Unlike static VLP systems, [21] provides the performance limits of dynamic active VLP by considering target mobility, offering insights into the nature of localization cooperation and long-term performance. [22] continues to study the CRLB of RSS-based SPO algorithms, acknowledging the potential absence of prior information and suggesting that mature performance analyses need to account for special cases such as positional inaccuracies. In [23], a CRLB based on multiple PDs collecting positioning signals from a single LED is derived, similar to the approach in [18]. Similarly, [24] derives a CRLB for localization using multiple PDs, displaying that cooperation between PDs can remarkably enhance VLP performance. A method based on RSS without model calibration is proposed in [25], though it lacks an in-depth analysis of the CRLB's analytical solution. It is worth emphasizing that all the aforementioned studies focus on active VLP systems. The recent study on the theoretical performance bounds of passive VLP, covered only in [17], demonstrates that further principle analyses are yet to be explored in depth.

Overall, despite significant efforts, much remains to be understood about the performance bounds of RSS-based passive VLP, particularly when accounting for a moving target.

B. Contributions of This Paper

The principles of positioning error evolution, the conditions for localization error convergence, the intrinsic mechanisms of positioning collaboration, and the impact of dominant factors on the performance of RSS-based passive VLP schemes have not been fully clarified at a theoretical level. Additionally, the intrinsic principle by which measurement noise with randomness influences positioning error evolution needs to be deeply explored. Considering the practical scenario where a PD, as a sensor node, may have errors in initial position acquisition, it is of great practical significance to examine how PD position uncertainty—a key factor affecting VLP errors—manifests in the performance limits. To accomplish a good balance between localization accuracy and deployment cost, it is also essential to quantitatively learn the specific scaling relationship between localization error and the number of LEDs or sensors required. Driven by the above research motivations, this paper aims to investigate the performance bounds of a passive VLP system based on the RSS measurements and statistical modeling. We focus on the principles of error evolution for both mobile target positioning and PD location calibration, revealing the relevant asymptotic performance limits of the contributing factors. This paper specifically attempts to answer the following questions.

- Given the system parameters, channel model and target mobility, what performance can the RSS-based passive VLP system achieve at each time slot?
- How do the positioning and calibration errors of this VLP system evolve over time, and how are long-term performance limits of this passive system determined?
- How do the factors such as prior knowledge, measurement noise, sensor location error and device deployment density affect the VLP performance of this passive system?

These questions will be addressed through the performance analysis of the closed-form CRLBs, which is challenging due to the need of extracting parametric information about target positions from complex channel functions. The contributions of this paper are mainly threefold as follows.

- *Closed-Form CRLB on Error:* We derive the performance bounds of this passive VLP system, quantified by the closed-form CRLBs for instantaneous target localization error and PD position calibration error. This analysis distinctly demonstrates the informative contributions of real-time observation data and prior prediction information to the instantaneous performance.
- *Closed-Form Analysis on Error Evolution:* We reveal the qualitatively analyzed principles of error evolution for target localization and PD calibration. Subsequently, we examine the

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convergence conditions and properties of this error evolution, and analyze the limits of long-term VLP performance using the closed-form steady-state solution of the bound. Additionally, we elucidate the information exchange mechanism between moving target localization and sensor position calibration.

• *Sophisticated Qualitative Performance Analysis on Error:* Apart from traditional quantitative numerical simulations, we focus on qualitatively analyzing asymptotic performance limits. This analysis reveals the specific effects of crucial factors such as prior knowledge, measurement noise, sensor location error and device deployment density on error evolution, giving constructive insights into the design of robust VLP algorithms.

It is crucial to emphasize that the PD sensors in this system are widely distributed throughout the environment. These PDs can be positioned on the floor or on surrounding walls. For the sake of modeling, this paper focuses on a scenario where the PDs are located on the floor. However, the proposed method is extendable to cases where the PDs are positioned elsewhere in the environment by modifying the shadow occlusion model, showing that the analytical framework remains generalizable.

The rest of this paper is organized as follows. In Section II, the system description is provided to derive the errors. Section III derives the closed-form CRLBs to study the error evolution for both target localization and PD position calibration. Section IV reveals the long-term VLP performance by analyzing the error evolution convergence conditions and properties. Section V shows the asymptotic performance analysis. The simulation results are presented in Section VI. Finally, what we have done in this work is concluded in Section VII. The symbol notations for crucial physical parameters are summarized in Table I.

TABLE I
OPERATION SYMBOL NOTATIONS

$\mathbf{x}_t$	target position at time t
$\boldsymbol{\chi}$	precision matrix related to $\mathbf{x}_t$
$\mathbf{r}_t^i$	true position of the ith PD at time t
$\mathbf{r}_t$	PD variable vector used for target localization
$\mathbf{s}_t^i$	false position of the ith PD at time t
$\mathbf{U}_t^i$	precision matrix related to $\mathbf{r}_t^i$
$\mathbf{t}_t^j$	true position of the jth LED at time t
$\mathbf{t}_t$	LED variable vector used for sensor calibration
$\mathbf{u}^i$	orientation vector of the ith PD
$\mathbf{v}^j$	orientation vector of the jth LED
$\rho_t^{i,j}$	transmission distance between the ith PD and the jth LED
$\varrho_t^{i,j}$	transmission distance between the target and the jth LED
$\mathbf{e}_{\mathbf{r},t}^{i,j}$	incidence vector determined by the the ith PD and the jth LED
$\mathbf{e}_{\mathbf{x},t}^{i,j}$	incidence vector determined by the target and the jth LED
$\theta_t^{i,j}$	incidence angle between $-\mathbf{e}_{\mathbf{r},t}^{i,j}$ and $\mathbf{u}^i$
$\phi_t^{i,j}$	irradiation angle between $\mathbf{e}_{\mathbf{x},t}^{i,j}$ and $\mathbf{v}^j$
$\theta_t^{j,i}$	incidence angle between $-\mathbf{e}_{\mathbf{r},t}^{i,j}$ and $\mathbf{u}^i$
$\phi_t^{j,i}$	irradiation angle between $\mathbf{e}_{\mathbf{r},t}^{i,j}$ and $\mathbf{v}^j$
$z_t^{i,j}$	RSS measurement information used for target localization
$\mathbf{z}_t$	overall measurement information vector used for target localization
$\epsilon_t^{i,j}$	Gaussian noise used for target localization
$\omega_t^{i,j}$	noise precision of $\epsilon_t^{i,j}$
$\boldsymbol{\omega}_t$	noise precision vector used for target localization
$\boldsymbol{\Lambda}$	scaling matrice related to $\omega_t^{i,j}$
λ	degree of freedom related to $\omega_t^{i,j}$
Ω_t^j	set of positioning PDs used for target localization
Ω_t	size of Ω_t^j used for target localization
$\boldsymbol{\alpha}_t$	complete variable related to target localization
$\boldsymbol{\vartheta}$	signal-noise ratio related to target localization
$f_t^{j,i}$	RSS measurement information used for sensor calibration
$\mathbf{f}_t$	overall measurement information vector used for sensor calibration
$\varepsilon_t^{j,i}$	Gaussian noise used for sensor calibration
$\psi_t^{j,i}$	noise precision of $\varepsilon_t^{j,i}$
$\boldsymbol{\psi}_t$	noise precision vector used for sensor calibration
$\boldsymbol{\Gamma}$	scaling matrice related to $\psi_t^{j,i}$
γ	degree of freedom related to $\psi_t^{j,i}$
Φ_t^i	set of effective LEDs used for sensor calibration of $\mathbf{r}_t^i$
Φ_t	size of Φ_t^i used for sensor calibration of $\mathbf{r}_t^i$
$\boldsymbol{\beta}_t$	complete variable retated to sensor calibration
$\boldsymbol{\varpi}$	signal-noise ratio related to sensor calibration

II. SYSTEM DESCRIPTION

In this section, we elaborate on the architecture of a passive VLP system, explicate the measurement model applied for the specific localization and calibration implementations, and also establish a target movement model. Based on these models, we provide a statistical framework for deriving error bounds.

A. System Model

We consider a passive VLP system with M LED transmitters and N PD sensors for real-time tracking and positioning of a target moving randomly within a room, as shown in Fig. 1. We assume that all PDs are randomly and uniformly deployed on the room's floor. However, owing to inevitable positional bias during deployment, the initial positions of these PDs may not be completely accurate. Meanwhile, the LEDs are evenly mounted on the room's ceiling with reasonable deployment intervals to minimize interference, without compromising the collaborative localization of these LEDs [26]. Let $\mathbf{x}_t \in \mathbb{R}^3$ represent the estimated target position at time t. The true positions of the ith PD and the jth LED required for locating the moving target at time t are denoted by $\mathbf{r}_t^i \in \mathbb{R}^3$ and $\mathbf{t}_t^j \in \mathbb{R}^3$, respectively. Note that all PD sensor positions are unknown due to deployment errors, whereas all LED locations are known and accurate. In addition, we assume that all PDs and LEDs here are vertically oriented, with $\mathbf{u}^i = [0, 0, 1]^\top$ and $\mathbf{v}^j = [0, 0, -1]^\top$ being the orientation vectors of the ith PD and the jth LED, respectively, where $\bullet^\top$ denotes the matrix transpose. Moreover, the reference LED is the specific single light used for target localization, which is randomly selected from multiple LEDs. We also hypothesize that multiple access protocols are implemented in this passive VLP system [27]. This assumption ensures that different PDs can recognize the light signals from various LEDs, preventing potential signal interference that might be challenging to distinguish.

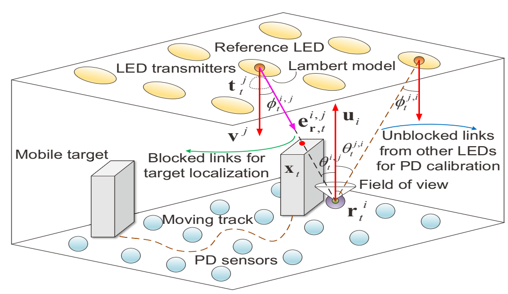

Fig. 1. Illustration of the RSS-based passive VLP system.

B. Measurement Model

The work [17] offers a paradigm for achieving localization by relying on a single LED transmitter to identify the set of PD sensors occluded by a localized target, providing a robust theoretical foundation for passive VLP performance in a multi-light scenario. Building on this, we continue to employ this single-light-based localization scheme to achieve optimal VLP performance. The selection of such a reference LED underlines that the target's position estimation at any moment depends on this preselected LED. The choice of this reference LED sets the upper limit of VLP performance at the initial moment [28].

We can assume that $\mathbf{r}_t^i$, $\mathbf{x}_t$ and $\mathbf{t}_t^j$ adhere to the principle of three-point covariance when the *i*th PD is fully blocked. This assumption ensures that an accurate estimate of the geometric center position of the target can be thus obtained. For notation ease, we let $\rho_t^{i,j} = ||\mathbf{r}_t^i - \mathbf{t}_t^j||_2$ and $\varrho_t^{i,j} = ||\mathbf{x}_t - \mathbf{t}_t^j||_2$ denote the transmission distances between the LED-PD and LED-target pair at time *t*, respectively, where $||\bullet||_2$ is the $\mathcal{L}_2$-norm of the vector. The incidence vector determined by the associated pair can be represented in two expressions: $\mathbf{e}_{\mathbf{x},t}^{i,j} = (\mathbf{x}_t - \mathbf{t}_t^j)/\varrho_t^{i,j} \in \mathbb{R}^3$ and $\mathbf{e}_{\mathbf{r},t}^{i,j} = (\mathbf{r}_t^i - \mathbf{t}_t^j)/\rho_t^{i,j} \in \mathbb{R}^3$. Additionally, let $\phi_t^{i,j} = \arccos((\mathbf{x}_t - \mathbf{t}^j)^\top \mathbf{v}^j/\varrho_t^{i,j})$ define the irradiation angle between $\mathbf{e}_{\mathbf{x},t}^{i,j}$ and $\mathbf{v}^j$, and let $\theta_t^{i,j} = \arccos((\mathbf{t}^j - \mathbf{r}_t^i)^\top \mathbf{u}^i/\rho_t^{i,j})$ denote the incidence angle between $-\mathbf{e}_{\mathbf{r},t}^{i,j}$ and $\mathbf{u}^i$. We use $\mathrm{z}_{t,e}^{i,j}$ and $\mathrm{z}_{t,o}^{i,j}$ to represent the RSS measurements from the *j*th LED received by the *i*th PD before and after the target enters this room. The RSS measurement difference, denoted as $\mathrm{z}_t^{i,j} = \mathrm{z}_{t,e}^{i,j} - \mathrm{z}_{t,o}^{i,j}$, is nonzero only when the *j*th PD is jammed by the target at time *t*. Under the assumption that the measurement difference can be considered as a superposition of line-of-sight (LOS) signal and Gaussian noise, we effectively mitigate the interference of non-line-of-sight (NLOS) signals on VLP performance by modeling RSS measurement differences [29]. Therefore, the crucial information source $\mathrm{z}_t^{i,j} \in \mathbb{R}$ used for target localization can be defined as the following form of [30]

$$\mathrm{z}_t^{i,j} = h(\mathbf{x}_t, \mathbf{r}_t^i, \mathbf{t}^j) + \epsilon_t^{i,j}, \forall i \in \Omega_t^j, \tag{1}$$

where $\epsilon_t^{i,j}$ as a zero-mean Gaussian noise with inverse variance $\omega_t^{i,j}$ denoting the measurement error can be supposed as $\epsilon_t^{i,j} \sim \mathcal{N}(0, \omega_t^{i,j})$ [31], and Ω_t^j is the set of blocked positioning PDs at time *t*, given as $\Omega_t^j = \{i|\theta_t^{i,j} \leqslant \theta_{\text{fov}}, \phi_t^{i,j} \leqslant \phi_{\text{fov}}, \mathrm{z}_t^{i,j} > 0, \forall i = 1:N, \exists j \in [1, M]\}$, where θ_{fov} and ϕ_{fov} are the field of view (FOV) of the PD and the LED, respectively, commonly assumed to be $\pi/2$ [32]. In addition, the RSS measurement function $h(\mathbf{x}_t, \mathbf{r}_t^i, \mathbf{t}^j)$ is defined by the LOS channel as [33]

$$h(\mathbf{x}_t, \mathbf{r}_t^i, \mathbf{t}^j) = -\Psi_R(d+1)(\varrho_t^{i,j})^{-2}\cos^d(\phi_t^{i,j})\cos(\theta_t^{i,j}), \tag{2}$$

where d denotes the Lambertian order that characterizes the LED radiation pattern [34], and $\Psi_R = P_T A_R F_R C_R/(2\pi)$ is a constant determined by the inherent physical parameters of the reference LED and the occluded PD [35]. Here, P_T represents the emitting power of the LED, A_R is the photosensitive area of the PD, F_R defines the optical filter gain of the PD, and C_R denotes the optical concentrator gain of the PD.

Then, consider the specific relationship between $\{\phi_t^{i,j}, \theta_t^{i,j}\}$ and $\{\mathbf{x}_t, \mathbf{r}_t^i, \mathbf{t}^j\}$, (2) can be rewritten as follows,

$$h(\mathbf{x}_t, \mathbf{r}_t^i, \mathbf{t}^j) = -\Psi_R \frac{(d+1)((\mathbf{x}_t - \mathbf{t}^j)^\top \mathbf{v}^j)^d (\mathbf{r}_t^i - \mathbf{t}^j)^\top \mathbf{u}^i}{||\mathbf{x}_t - \mathbf{t}^j||_2^d \, ||\mathbf{r}_t^i - \mathbf{t}^j||_2^3}. \tag{3}$$

Given the challenge of accurately determining the unknown fine location of the obstructed PD, denoted by $\mathbf{r}_t^i$ in (3), it may prove beneficial to employ the coarse location of the PD, defined by $\mathbf{s}_t^i$, to establish a connection between the true and false locations. Generally, Gaussian models are widely used and deemed valid assumptions to mitigate modeling estimation errors [36]. Then, $\mathbf{r}_t^i$ can be modeled as a Gaussian distribution variable, characterized by the mean vector $\mathbf{s}_t^i \in \mathbb{R}^3$ and the precision matrix $\mathbf{U}_t^i \in \mathbb{S}^3$ [37]. Namely, $\mathbf{r}_t^i \sim \mathcal{N}(\mathbf{s}_t^i, \mathbf{U}_t^i), \forall i = 1:\Omega_t$, where $\mathbf{U}_t^i \to \boldsymbol{\infty}$ indicates that the location is entirely accurate, and Ω_t represents the size of the blocked PD set Ω_t^j.

While calibrating the locations of positioning PDs is theoretically critical for real-time tracking and target localization, we do not view the calibration of PDs and target localization as strictly separate or sequential stages. Specifically, when the moving target obstructs the *i*th PD under the influence of the *j*th reference LED at time *t*, this PD is identified as requiring calibration. By using LOS signals from LEDs other than the *j*th LED, which illuminate the PD indirectly, and integrating the single link provided by the *j*th LED during target localization, we attain an active calibration strategy. This scheme leverages target localization data to assist in active PD sensor calibration, making localization and calibration complementary processes. Besides, to emphasize the informative contribution of visible light to sensor position calibration, we assume the absence of localization collaboration among PDs, indicating no possibility of information transfer between them.

Likewise, we can denote the irradiation angle between $\mathbf{e}_{\mathbf{r},t}^{i,j}$ and $\mathbf{v}^j$ as $\phi_t^{j,i} = \arccos((\mathbf{r}_t^i - \mathbf{t}_t^j)^\top \mathbf{v}^j/\varrho_t^{i,j})$, and let $\theta_t^{j,i} = \arccos((\mathbf{t}_t^j - \mathbf{r}_t^i)^\top \mathbf{u}^i/\rho_t^{i,j})$ be the incidence angle between $\mathbf{u}^i$ and $-\mathbf{e}_{\mathbf{r},t}^{i,j}$. The RSS measurement $\mathrm{f}_t^{j,i} \in \mathbb{R}$ between the *i*th PD and the *j*th LED, which offers the LOS path, is given in

$$\mathrm{f}_t^{j,i} = l(\mathbf{t}_t^j, \mathbf{r}_t^i) + \varepsilon_t^{j,i}, \forall j \in \Phi_t^i, \tag{4}$$

where $\varepsilon_t^{j,i}$ is the noise component with precision $\psi_t^{j,i}$ defined as $\varepsilon_t^{j,i} \sim \mathcal{N}(0, \psi_t^{j,i})$, while Φ_t^i is the set of effective LEDs for calibrating $\mathbf{r}_t^i$, which can be expressed in the form of $\Phi_t^i = \{j|\theta_t^{j,i} \leqslant \theta_{\text{fov}}, \phi_t^{j,i} \leqslant \phi_{\text{fov}}, \mathrm{z}_t^{i,j} = 0, \forall j = 1:M, \exists i \in \Omega_t^j\}$, and $l(\mathbf{t}_t^j, \mathbf{r}_t^i)$ denotes the relevant RSS measurement function after substituting $\phi_t^{j,i}$ and $\theta_t^{j,i}$ into (2), given by

$$l(\mathbf{t}_t^j, \mathbf{r}_t^i) = -\Psi_R \frac{(d+1)((\mathbf{r}_t^i - \mathbf{t}_t^j)^\top \mathbf{v}^j)^d (\mathbf{r}_t^i - \mathbf{t}_t^j)^\top \mathbf{u}^i}{||\mathbf{r}_t^i - \mathbf{t}_t^j||_2^{d+3}}. \tag{5}$$

As the precision of measurement information can be influenced by various factors in the temporal and spatial domains, it is essential to effectively assess the variability of different measurement information captured by various PDs at different time points. To accomplish this, we propose introducing the Wishart distribution [38], a common method for evaluating the uncertainty of accuracy, to model each noise precision, i.e.,

$$\omega_t^{i,j} \sim \mathcal{W}(\boldsymbol{\Lambda}, \lambda), \forall i = 1:\Omega_t \; ; \; \psi_t^{j,i} \sim \mathcal{W}(\boldsymbol{\Gamma}, \gamma), \forall j = 1:\Phi_t, \tag{6}$$

in which both $\boldsymbol{\Lambda}$ and $\boldsymbol{\Gamma}$ denote the respective scaling matrices, both λ and γ mean the associated degrees of freedom (DoF). When the noise precision satisfies a one-dimensional central Wishart distribution, it will degenerate to a Chi-square random variable [39]. At this point, the scaling matrices essentially are the corresponding variances. Additionally, Φ_t is the size of the LED set Φ_t^i for PD position calibration. Then, we can denote $\boldsymbol{\omega}_t = [\omega_t^{1,j}; \cdots; \omega_t^{\Omega_t,j}] \in \mathbb{R}^{\Omega_t}$ as the noise accuracy vector for target localization, and similarly, let $\boldsymbol{\psi}_t = [\psi_t^{1,i}; \cdots; \psi_t^{\Phi_t,i}] \in \mathbb{R}^{\Phi_t}$ be the noise precision vector for PD position calibration. For the sake of simplicity, Ω_t and Φ_t are collectively referred to as the concept of reference set sizes.

Equally, let $\mathbf{r}_t = [\mathbf{r}_t^1; \cdots; \mathbf{r}_t^{\Omega_t}] \in \mathbb{R}^{3\Omega_t}$ be the PD variable vector for localization, and denote $\mathbf{t}_t = [\mathbf{t}_t^1; \cdots; \mathbf{t}_t^{\Phi_t}] \in \mathbb{R}^{3\Phi_t}$ as the LED variable vector for calibration. Besides, $\mathrm{z}_t^{i,j}$ in (1) can be stacked in the form of $\mathbf{z}_t = [\mathrm{z}_t^{1,j}; \cdots; \mathrm{z}_t^{\Omega_t,j}] \in \mathbb{R}^{\Omega_t}$, and $\mathbf{f}_t = [\mathrm{z}_t^{i,j}; \mathrm{f}_t^{1,i}; \cdots; \mathrm{f}_t^{\Phi_t,i}] \in \mathbb{R}^{\Phi_t+1}$ can represent the overall measurement information vector required for calibrating $\mathbf{r}_t^i$.

Overall, this system is based on two main components: (i) $\{\mathrm{z}_t^{i,j}, \mathbf{s}_t^i, \mathbf{U}_t^i : \forall i \in \Omega_t^j\}$, the current RSS observation and the rough position of the occluded PD adjusted by error accuracy, to track and localize the moving target; and (ii) $\{\mathrm{f}_t^{j,i}, \mathbf{t}_t^j : \forall j \in \Phi_t^i; \mathrm{z}_t^{i,j}\}$, the RSS measurement provided by other LEDs along with the observation related to the mobile target, to facilitate the required position calibration of each inaccurate sensor $\mathbf{r}_t^i$.

C. Mobility Model

To construct a generalized performance analysis framework solely for VLP, we assume the complete uncertainty regarding the target's movement direction at any given moment. We do not consider the results of other localization methods as prior knowledge for enhancing VLP performance. We assume that the current target position $\mathbf{x}_t$ moves from its previous location $\mathbf{x}_{t-1} \in \mathbb{R}^3$ according to a transition vector $\boldsymbol{\tau}_t \in \mathbb{R}^3$ [40], i.e., $\mathbf{x}_t = \mathbf{x}_{t-1} + \boldsymbol{\tau}_t$. Here, $\boldsymbol{\tau}_t$, defined as $\boldsymbol{\tau}_t \sim \mathcal{N}(\mathbf{0}_{3\times 1}, \boldsymbol{\chi})$, is a zero-mean Gaussian variable with a precision matrix $\boldsymbol{\chi} \in \mathbb{S}^3$, which is widely used in mobility modeling. Thus, $\mathbf{x}_t$ can be expressed as $\mathbf{x}_t \sim \mathcal{N}(\mathbf{x}_{t-1}, \boldsymbol{\chi})$, effectively linking the target's position after movement to the transfer accuracy.

For simplicity, we assume that the position transition precision, reflecting the target mobility, remains constant over time. A larger value of $\boldsymbol{\chi}$ implies a smaller degree of position shift, meaning that the target positions at two consecutive moments are closer to each other. Although more complex models exist for characterizing target movement [41], this paper focuses exclusively on the mobility model described above to facilitate subsequent systematic statistical modeling and analysis.

D. Statistical Model for Localization and Calibration

Let $\boldsymbol{\alpha}_t = [\mathbf{x}_t; \mathbf{t}^j; \mathbf{r}_t; \boldsymbol{\omega}_t] \in \mathbb{R}^{6+4\Omega_t}$ as a complete variable related to target localization. As per (1), the likelihood function $p(\mathbf{z}_t|\boldsymbol{\alpha}_t)$ obeying $\mathrm{z}_t^{i,j} \sim \mathcal{N}(h(\mathbf{x}_t, \mathbf{r}_t^i, \mathbf{t}^j), \omega_t^{i,j})$ is given by

$$p(\mathbf{z}_t|\boldsymbol{\alpha}_t) = \prod_{i\in\Omega_t^j} \mathcal{N}(\mathrm{z}_t^{i,j}|h(\mathbf{x}_t, \mathbf{r}_t^i, \mathbf{t}^j), \omega_t^{i,j}). \tag{7}$$

The state transition distribution is defined as $p(\boldsymbol{\alpha}_t|\boldsymbol{\alpha}_{t-1}) = \mathcal{N}(\mathbf{x}_t|\mathbf{x}_{t-1}, \boldsymbol{\chi}) \prod_{i\in\Omega_t^j} \mathcal{N}(\mathbf{r}_t^i|\mathbf{s}_t^i, \mathbf{U}_t^i)\mathcal{W}(\omega_t^{i,j}|\boldsymbol{\Lambda}, \lambda)$, where notation $\mathcal{W}(\omega_t^{i,j}|\boldsymbol{\Lambda}, \lambda)$ means $\omega_t^{i,j} \sim \mathcal{W}(\boldsymbol{\Lambda}, \lambda)$. Bayesian statistics theory can well be effectively applied to the statistical model analysis when the localization information from the previous moment is considered as prior prediction information for the current moment's localization [42]. Under this condition, the estimation error tends to follow a Gaussian distribution when the sample size is sufficiently large. Consequently, the previous posterior distribution $p(\mathbf{x}_{t-1}|\mathbf{z}_{1:t-1})$ can be asymptotically approximated by a Gaussian distribution as $p(\mathbf{x}_{t-1}|\mathbf{z}_{1:t-1}) \approx \mathcal{N}(\mathbf{x}_{t-1}|\mathbf{x}_{t-1}^\flat, \boldsymbol{\mathcal{H}}_{\mathrm{BE}}^{\mathbf{x}_{t-1}})$, where $\mathbf{z}_{1:t-1} = \{\mathbf{z}_1, \cdots, \mathbf{z}_{t-1}\}$, while $\mathbf{x}_{t-1}^\flat \in \mathbb{R}^3$ is the previous posterior location estimate, and $\boldsymbol{\mathcal{H}}_{\mathrm{BE}}^{\mathbf{x}_{t-1}} \in \mathbb{S}^3$ is the Fisher information matrix (FIM) based on Bayesian estimation (BE) in the previous posterior state. Thus, the prediction distribution of $\boldsymbol{\alpha}_t$ can be derived as

$$\begin{aligned} &p(\boldsymbol{\alpha}_t|\mathbf{z}_{1:t-1}) \\ &= \int p(\boldsymbol{\alpha}_t|\boldsymbol{\alpha}_{t-1})p(\boldsymbol{\alpha}_{t-1}|\mathbf{z}_{1:t-1})d\boldsymbol{\alpha}_{t-1} \\ &= \int p(\mathbf{x}_t|\mathbf{x}_{t-1})p(\mathbf{x}_{t-1}|\mathbf{z}_{1:t-1})d\mathbf{x}_{t-1}\int p(\mathbf{r}_t)d\mathbf{r}_t\int p(\boldsymbol{\omega}_t^1)d\boldsymbol{\omega}_t^1 \\ &\approx \int \mathcal{N}(\mathbf{x}_t|\mathbf{x}_{t-1}, \boldsymbol{\chi})\mathcal{N}(\mathbf{x}_{t-1}|\mathbf{x}_{t-1}^\flat, \boldsymbol{\mathcal{H}}_{\mathrm{BE}}^{\mathbf{x}_{t-1}})d\mathbf{x}_{t-1}\prod_{i\in\Omega_t^j} p(\mathbf{r}_t^i)p(\omega_t^{i,j}) \\ &= \mathcal{N}(\mathbf{x}_t|\mathbf{x}_{t-1}^\flat, \boldsymbol{\Xi}_t^{\mathrm{pre}}) \prod_{i\in\Omega_t^j} \mathcal{N}(\mathbf{r}_t^i|\mathbf{s}_t^i, \mathbf{U}_t^i)\mathcal{W}(\omega_t^{i,j}|\boldsymbol{\Lambda}, \lambda), \end{aligned} \tag{8}$$

where $\boldsymbol{\Xi}_t^{\mathrm{pre}} \in \mathbb{S}^3$ is defined as $\boldsymbol{\Xi}_t^{\mathrm{pre}} = (\boldsymbol{\chi}^{-1} + (\boldsymbol{\mathcal{H}}_{\mathrm{BE}}^{\mathbf{x}_{t-1}})^{-1})^{-1}$. By (8), the previous measurements $\mathbf{z}_{1:t-1}$ can be divided into two components: history information carried solely by $\mathbf{x}_t$, and priori prediction data jointly implied by $\{\mathbf{s}_t^i, \omega_t^{i,j} : \forall i \in \Omega_t^j\}$. As $p(\mathbf{z}_t|\mathbf{z}_{1:t-1})$ is a normalized constant, then the posterior distribution of $\boldsymbol{\alpha}_t$, i.e., $p(\boldsymbol{\alpha}_t|\mathbf{z}_{1:t})$, can be formed as

$$p(\boldsymbol{\alpha}_t|\mathbf{z}_{1:t}) = \frac{p(\mathbf{z}_t|\boldsymbol{\alpha}_t)p(\boldsymbol{\alpha}_t|\mathbf{z}_{1:t-1})}{p(\mathbf{z}_t|\mathbf{z}_{1:t-1})} \propto p(\mathbf{z}_t|\boldsymbol{\alpha}_t)p(\boldsymbol{\alpha}_t|\mathbf{z}_{1:t-1}), \tag{9}$$

in which $\propto$ implies the left term is proportional to the right and $\mathbf{z}_{1:t} = \{\mathbf{z}_1, \cdots, \mathbf{z}_t\}$, while (9), derived from (7) and (8), contains all the essential data required for target localization.

Additionally, define $\boldsymbol{\beta}_t = [\mathbf{r}_t^i; \mathbf{x}_t; \mathbf{t}_t; \omega_t^{i,j}; \boldsymbol{\psi}_t] \in \mathbb{R}^{7+4\Phi_t}$ as a complete variable related to PD position calibration, then the likelihood function conditioned on $\boldsymbol{\beta}_t$ can be written as

$$p(\mathbf{f}_t|\boldsymbol{\beta}_t) = \mathcal{N}(\mathrm{z}_t^{i,j}|h(\mathbf{x}_t, \mathbf{r}_t^i, \mathbf{t}^j), \omega_t^{i,j}) \prod_{j\in\Phi_t^i} \mathcal{N}(\mathrm{f}_t^{j,i}|l(\mathbf{t}_t^j, \mathbf{r}_t^i), \psi_t^{j,i}), \tag{10}$$

where $\mathrm{z}_t^{i,j}$ and $\{\mathrm{f}_t^{j,i} : \forall j \in \Phi_t^i\}$ are mutually independent. Let the marginal posterior distribution of the object position $\mathbf{x}_t$ denote as $p(\mathbf{x}_t|\mathbf{z}_t) = \iint p(\boldsymbol{\alpha}_t|\mathbf{z}_{1:t})d\mathbf{r}_t^i d\omega_t^{i,j}$, where $p(\boldsymbol{\alpha}_t|\mathbf{z}_{1:t})$ is given by (9). Similarly, we can also define $p(\boldsymbol{\beta}_t)$ as

$$p(\boldsymbol{\beta}_t) = p(\mathbf{x}_t|\mathbf{z}_t)\mathcal{N}(\mathbf{r}_t^i|\mathbf{s}_t^i, \mathbf{U}_t^i)\mathcal{W}(\omega_t^{i,j}|\boldsymbol{\Lambda}, \lambda)\prod_{j\in\Phi_t^i}\mathcal{W}(\psi_t^{j,i}|\boldsymbol{\Gamma}, \gamma). \tag{11}$$

Hence, based on the ideal constant $p(\mathbf{f}_t)$, (10) and (11), the posterior distribution of $\boldsymbol{\beta}_t$, i.e., $p(\boldsymbol{\beta}_t|\mathbf{f}_t)$, is formed as

$$p(\boldsymbol{\beta}_t|\mathbf{f}_t) = p(\mathbf{f}_t|\boldsymbol{\beta}_t)p(\boldsymbol{\beta}_t)/p(\mathbf{f}_t) \propto p(\mathbf{f}_t|\boldsymbol{\beta}_t)p(\boldsymbol{\beta}_t). \tag{12}$$

In general, (9) and (12) are the basis for CRLB analysis of subsequent localization and calibration issues.

III. Error Bound of VLP

In this section, we will reveal the closed-form CRLBs on the mobile target localization and the sensor position calibration estimation errors, respectively, at each time slot.

A. Target Localization Analysis

1) Closed-Form CRLB: The RSS-based passive VLP can be achieved via $\mathcal{P}_{\text{VLP}}: \hat{\boldsymbol{\alpha}}_t = \arg\max_{\boldsymbol{\alpha}_t} p(\boldsymbol{\alpha}_t|\mathbf{z}_{1:t})$, where $\hat{\mathbf{x}}_t$ in $\hat{\boldsymbol{\alpha}}_t$ is the estimate of $\mathbf{x}_t$. Based on Bayesian statistics theory [42], the location estimation error $||\hat{\mathbf{x}}_t - \mathbf{x}_t||_2$ is bounded by $\mathbb{E}_{\mathbf{z}_t}\{||\hat{\mathbf{x}}_t - \mathbf{x}_t||_2\} \geqslant \sqrt{\text{tr}(\boldsymbol{\mathcal{B}}_{\text{BE}}^{\mathbf{x}_t})}$, where $\mathbb{E}_{\mathbf{z}_t}\{\bullet\}$ denotes the expectation related to $p(\mathbf{z}_t)$, and $\text{tr}(\bullet)$ means the matrix trace, while $\boldsymbol{\mathcal{B}}_{\text{BE}}^{\mathbf{x}_t} \in \mathbb{S}^3$ indicates the closed-form BE-based CRLB on the position estimate error. A theorem will be presented to explain how this bound quantifies the informative contribution of observed and predicted information to VLP performance.

Theorem 1 (Single-Time-Slot Localization CRLB): At time t, the CRLB $\boldsymbol{\mathcal{B}}_{\text{BE}}^{\mathbf{x}_t}$ of target position is given by

$$\boldsymbol{\mathcal{B}}_{\text{BE}}^{\mathbf{x}_t} = \left(\boldsymbol{\vartheta} \cdot \boldsymbol{\mathcal{O}}_{\text{BE}}^{\mathbf{x}_t} + (\boldsymbol{\chi}^{-1} + \boldsymbol{\mathcal{B}}_{\text{BE}}^{\mathbf{x}_{t-1}})^{-1}\right)^{-1}, \tag{13}$$

where the signal-noise ratio (SNR) $\boldsymbol{\vartheta} \in \mathbb{R}$ is denoted as $\boldsymbol{\vartheta} = \Lambda\lambda\Psi_R^2$, $\boldsymbol{\mathcal{O}}_{\text{BE}}^{\mathbf{x}_t} \in \mathbb{S}^3$ as the equivalent measurement information is manifested by (14), thus we can define $\boldsymbol{\mathcal{L}}_{\text{obs}}^{\mathbf{x}_t} = \boldsymbol{\vartheta} \cdot \boldsymbol{\mathcal{O}}_{\text{BE}}^{\mathbf{x}_t} \in \mathbb{S}^3$ as the observation information from PDs. Similarly, let $\boldsymbol{\mathcal{I}}_{\text{pre}}^{\mathbf{x}_t} = \boldsymbol{\Xi}_t^{\text{pre}} = (\boldsymbol{\chi}^{-1} + \boldsymbol{\mathcal{B}}_{\text{BE}}^{\mathbf{x}_{t-1}})^{-1} \in \mathbb{S}^3$ be the prediction information from time recursion, where the previous CRLB is denoted as $\boldsymbol{\mathcal{B}}_{\text{BE}}^{\mathbf{x}_{t-1}} = (\boldsymbol{\mathcal{H}}_{\text{BE}}^{\mathbf{x}_{t-1}})^{-1} \in \mathbb{S}^3$. Hence, the BE-based FIM $\boldsymbol{\mathcal{H}}_{\text{BE}}^{\mathbf{x}_t} = \boldsymbol{\mathcal{L}}_{\text{obs}}^{\mathbf{x}_t} + \boldsymbol{\mathcal{I}}_{\text{pre}}^{\mathbf{x}_t} \in \mathbb{S}^3$ can be defined as the position information (accuracy), then $\boldsymbol{\mathcal{B}}_{\text{BE}}^{\mathbf{x}_t} = (\boldsymbol{\mathcal{H}}_{\text{BE}}^{\mathbf{x}_t})^{-1}$ holds.

Proof: Please see the proof in Appendix A.

Based on (13), it is evident that there is no prior parameter estimation at the initial moment. Therefore, it can be assumed that the FIM $\boldsymbol{\mathcal{H}}_{\text{BE}}^{\mathbf{x}_0} = \boldsymbol{\mathcal{L}}_{\text{obs}}^{\mathbf{x}_0}$ since $\boldsymbol{\mathcal{I}}_{\text{pre}}^{\mathbf{x}_0} = \mathbf{0}$, indicating that the related prediction information is invalid at this time. For ease of clarity, $\boldsymbol{\mathcal{O}}_{\text{BE}}^{\mathbf{x}_t}$ in Theorem 1 is explicated as follows, i.e.,

$$\boldsymbol{\mathcal{O}}_{\text{BE}}^{\mathbf{x}_t} = \mathbf{A}_{\natural\text{BE}}^{\mathbf{x}_t} - \mathbf{B}_{\natural\text{BE}}^{\mathbf{x}_t}(\mathbf{C}_{\natural\text{BE}}^{\mathbf{x}_t})^{-1}\mathbf{D}_{\natural\text{BE}}^{\mathbf{x}_t}, \tag{14}$$

where $\mathbf{A}_{\natural\text{BE}}^{\mathbf{x}_t}$, $\mathbf{B}_{\natural\text{BE}}^{\mathbf{x}_t}$, $\mathbf{C}_{\natural\text{BE}}^{\mathbf{x}_t}$ and $\mathbf{D}_{\natural\text{BE}}^{\mathbf{x}_t}$ are given, respectively, by

$$\begin{aligned}\mathbf{A}_{\natural\text{BE}}^{\mathbf{x}_t} \in \mathbb{R}^{3\times3} =& \mathbf{F}(\mathbf{x}_t)\mathbf{K}(\mathbf{u})\mathbf{K}(\mathbf{u})^\top\mathbf{F}(\mathbf{x}_t)^\top \\ &- \mathbf{M}(\mathbf{x}_t,\mathbf{r}_t)\mathbf{R}(\mathbf{r}_t)\mathbf{M}(\mathbf{x}_t,\mathbf{r}_t)^\top,\end{aligned} \tag{15}$$

$$\begin{aligned}\mathbf{B}_{\natural\text{BE}}^{\mathbf{x}_t} \in \mathbb{R}^{3\times3} =& \mathbf{F}(\mathbf{x}_t)\mathbf{K}(\mathbf{u})\mathbf{K}(\mathbf{u})^\top\mathbf{J}(\mathbf{t}^j)^\top \\ &- \mathbf{M}(\mathbf{x}_t,\mathbf{r}_t)\mathbf{R}(\mathbf{r}_t)\mathbf{N}(\mathbf{t}^j,\mathbf{r}_t)^\top,\end{aligned} \tag{16}$$

$$\begin{aligned}\mathbf{C}_{\natural\text{BE}}^{\mathbf{x}_t} \in \mathbb{R}^{3\times3} =& \mathbf{J}(\mathbf{t}^j)\mathbf{K}(\mathbf{u})\mathbf{K}(\mathbf{u})^\top\mathbf{J}(\mathbf{t}^j)^\top \\ &- \mathbf{N}(\mathbf{t}^j,\mathbf{r}_t)\mathbf{R}(\mathbf{r}_t)\mathbf{N}(\mathbf{t}^j,\mathbf{r}_t)^\top,\end{aligned} \tag{17}$$

$$\begin{aligned}\mathbf{D}_{\natural\text{BE}}^{\mathbf{x}_t} \in \mathbb{R}^{3\times3} =& \mathbf{J}(\mathbf{t}^j)\mathbf{K}(\mathbf{u})\mathbf{K}(\mathbf{u})^\top\mathbf{F}(\mathbf{x}_t)^\top \\ &- \mathbf{N}(\mathbf{t}^j,\mathbf{r}_t)\mathbf{R}(\mathbf{r}_t)\mathbf{M}(\mathbf{x}_t,\mathbf{r}_t)^\top,\end{aligned} \tag{18}$$

in which $\mathbf{F}(\mathbf{x}_t)$ and $\mathbf{K}(\mathbf{u})$ are defined, respectively, as

$$\mathbf{F}(\mathbf{x}_t) = \left[(\rho_t^{i,j})^{-2}\mathbf{f}^{i,j}(\mathbf{x}_t) \mid \forall i \in \Omega_t^j\right] \in \mathbb{R}^{3\times3\Omega_t}, \tag{19}$$

$$\mathbf{f}^{i,j}(\mathbf{x}_t) = \mathbf{f_1}^{i,j}(\mathbf{x}_t)\mathbf{f_2}^{i,j}(\mathbf{e}_{\mathbf{x},t}^{i,j}) \in \mathbb{R}^{3\times3}, \tag{20}$$

$$\mathbf{f_1}^{i,j}(\mathbf{x}_t) = (\varrho_t^{i,j})^{-1}(\mathbf{I}_3 - \mathbf{e}_{\mathbf{x},t}^{i,j}\mathbf{e}_{\mathbf{x},t}^{i,j\top}) \in \mathbb{R}^{3\times3}, \tag{21}$$

$$\begin{aligned}\mathbf{f_2}^{i,j}(\mathbf{e}_{\mathbf{x},t}^{i,j}) \in \mathbb{R}^{3\times3} =& d(d+1)(\mathbf{e}_{\mathbf{x},t}^{i,j\top}\mathbf{v}^j)^{d-1}\mathbf{v}^j\mathbf{e}_{\mathbf{x},t}^{i,j\top} \\ &+ (d+1)(\mathbf{e}_{\mathbf{x},t}^{i,j\top}\mathbf{v}^j)^d\mathbf{I}_3,\end{aligned} \tag{22}$$

$$\mathbf{K}(\mathbf{u}) = \mathbf{I}_{\Omega_t} \otimes \mathbf{u}^i \in \mathbb{R}^{3\Omega_t\times\Omega_t}, \tag{23}$$

while $\mathbf{M}(\mathbf{x}_t,\mathbf{r}_t)$ is denoted by the following equations as

$$\mathbf{M}(\mathbf{x}_t,\mathbf{r}_t) = \left[\mathbf{f}(\mathbf{x}_t)\mathbf{u}^i\mathbf{u}^{i\top}\mathbf{q}(\mathbf{r}_t^i)^\top \mid \forall i \in \Omega_t^j\right] \in \mathbb{R}^{3\times3\Omega_t}, \tag{24}$$

$$\mathbf{f}(\mathbf{x}_t) = (\rho_t^{i,j})^{-2}\mathbf{f}^{i,j}(\mathbf{x}_t) \in \mathbb{R}^{3\times3}, \tag{25}$$

$$\mathbf{q}(\mathbf{r}_t^i) = (\rho_t^{i,j})^{-2}\mathbf{q}^{i,j}(\mathbf{r}_t^i) \in \mathbb{R}^{3\times3}, \tag{26}$$

$$\mathbf{q}^{i,j}(\mathbf{r}_t^i) = \mathbf{f_5}^{i,j}(\mathbf{r}_t^i) - 2(\rho_t^{i,j})^{-2}(\mathbf{r}_t^i - \mathbf{t}^j)\mathbf{g}_t^{i,j}(\mathbf{e}_{\mathbf{r},t}^{i,j})^\top, \tag{27}$$

$$\mathbf{f_5}^{i,j}(\mathbf{r}_t^i) = \mathbf{f_3}^{i,j}(\mathbf{r}_t^i)\mathbf{f_4}^{i,j}(\mathbf{e}_{\mathbf{r},t}^{i,j}) \in \mathbb{R}^{3\times3}, \tag{28}$$

$$\mathbf{f_3}^{i,j}(\mathbf{r}_t^i) = (\rho_t^{i,j})^{-1}(\mathbf{I}_3 - \mathbf{e}_{\mathbf{r},t}^{i,j}\mathbf{e}_{\mathbf{r},t}^{i,j\top}) \in \mathbb{R}^{3\times3}, \tag{29}$$

$$\begin{aligned}\mathbf{f_4}^{i,j}(\mathbf{e}_{\mathbf{r},t}^{i,j}) \in \mathbb{R}^{3\times3} =& d(d+1)(\mathbf{e}_{\mathbf{r},t}^{i,j\top}\mathbf{v}^j)^{d-1}\mathbf{v}^j\mathbf{e}_{\mathbf{r},t}^{i,j\top} \\ &+ (d+1)(\mathbf{e}_{\mathbf{r},t}^{i,j\top}\mathbf{v}^j)^d\mathbf{I}_3,\end{aligned} \tag{30}$$

$$\mathbf{g}_t^{i,j}(\mathbf{e}_{\mathbf{r},t}^{i,j}) = (d+1)(\mathbf{e}_{\mathbf{r},t}^{i,j\top}\mathbf{v}^j)^d\mathbf{e}_{\mathbf{r},t}^{i,j} \in \mathbb{R}^3, \tag{31}$$

and $\mathbf{R}(\mathbf{r}_t)$, $\mathbf{J}(\mathbf{t}^j)$ and $\mathbf{N}(\mathbf{t}^j,\mathbf{r}_t)$ are given, respectively, by

$$\mathbf{R}(\mathbf{r}_t) = diag\left(\mathbf{r}(\mathbf{r}_t^1),\cdots,\mathbf{r}(\mathbf{r}_t^{\Omega_t})\right) \in \mathbb{R}^{3\Omega_t\times3\Omega_t}, \tag{32}$$

$$\mathbf{r}(\mathbf{r}_t^i) = \left(\mathbf{q}(\mathbf{r}_t^i)\mathbf{u}^i\mathbf{u}^{i\top}\mathbf{q}(\mathbf{r}_t^i)^\top + \text{SNR}^{-1}\mathbf{U}_t^i\right)^{-1} \in \mathbb{R}^{3\times3}, \tag{33}$$

$$\mathbf{J}(\mathbf{t}^j) = \left[(\rho_t^{i,j})^{-2}\mathbf{j}^{i,j}(\mathbf{t}^j) \mid \forall i \in \Omega_t^j\right] \in \mathbb{R}^{3\times3\Omega_t}, \tag{34}$$

$$\mathbf{j}^{i,j}(\mathbf{t}^j) = -\mathbf{f_5}^{i,j}(\mathbf{r}_t^i) - 2(\rho_t^{i,j})^{-2}(\mathbf{t}^j - \mathbf{r}_t^i)\mathbf{g}_t^{i,j}(\mathbf{e}_{\mathbf{r},t}^{i,j})^\top, \tag{35}$$

$$\mathbf{N}(\mathbf{t}^j,\mathbf{r}_t) = \left[\mathbf{j}(\mathbf{t}^j)\mathbf{u}^i\mathbf{u}^{i\top}\mathbf{q}(\mathbf{r}_t^i)^\top \mid \forall i \in \Omega_t^j\right] \in \mathbb{R}^{3\times3\Omega_t}, \tag{36}$$

$$\mathbf{j}(\mathbf{t}^j) = (\rho_t^{i,j})^{-2}\mathbf{j}^{i,j}(\mathbf{t}^j) \in \mathbb{R}^{3\times3}, \tag{37}$$

where in the equations mentioned above, specific symbols have distinct meanings: $[\bullet]$ in (19), (24), (34) and (36) denotes the stacking of elements in rows; $\mathbf{I}_3$ in (21), (22), (29) and (30) represents a three-dimensional unit matrix; $\mathbf{I}_{\Omega_t}$ in (23) similarly signifies a Ω_t-dimensional identity matrix; $\otimes$ in (23) is the Kronecker product operation; $diag(\bullet)$ in (32) means the generation of a diagonal matrix. Besides, by (27) and (35), we have $\mathbf{j}^{i,j}(\mathbf{t}^j) = -\mathbf{q}^{i,j}(\mathbf{r}_t^i)$, thus we know $\mathbf{j}(\mathbf{t}^j) = -\mathbf{q}(\mathbf{r}_t^i)$. By combining the four key important sub-conclusions of (15)–(18), the value of $\boldsymbol{\mathcal{O}}_{\text{BE}}^{\mathbf{x}_t}$ shown in (14) can be easily obtained.

2) Comments on Error Limit: Based on Theorem 1, we derive the following remarks on the manifestation and formation of error bounds for target position estimation at each moment.

Remark 1: From (13), it is evident that the total position information within this VLP architecture, denoted as $\boldsymbol{\mathcal{H}}_{\text{BE}}^{\mathbf{x}_t}$, encompasses both the RSS observation data and the movement prediction information. This combination is essential for assessing VLP accuracy, inversely represented by the CRLB $\boldsymbol{\mathcal{B}}_{\text{BE}}^{\mathbf{x}_t}$. The overall prediction information at the current moment $\boldsymbol{\mathcal{I}}_{\text{pre}}^{\mathbf{x}_t}$ is temporally recursive, depending on the target mobility characterized by the diversion variance matrix $\boldsymbol{\chi}^{-1}$ and the VLP error $\boldsymbol{\mathcal{B}}_{\text{BE}}^{\mathbf{x}_{t-1}}$ from the previous moment. Thus, apart from the dominant RSS observation information $\boldsymbol{\mathcal{L}}_{\text{obs}}^{\mathbf{x}_t}$ provided by the set of occluded PDs, both the previous VLP performance and the inherent target mobility can significantly influence the current VLP performance. The information transfer and the VLP error evolution from $\mathcal{B}_{\text{BE}}^{\mathbf{x}_{t-1}}$ to $\mathcal{B}_{\text{BE}}^{\mathbf{x}_t}$ imply that mobile target localization involves the time-domain propagation of positioning information. Consequently, this phenomenon undoubtedly impacts the long-term VLP performance, which will be elaborated upon through a detailed Section IV.

Remark 2: (13) shows that $\mathcal{L}_{\text{obs}}^{\mathbf{x}_t}$ comes from the equivalent measurement information $\mathcal{O}_{\text{BE}}^{\mathbf{x}_t}$ and the variable ϑ measuring the noise intensity. The inverse of ϑ reflects the extent to which noise components obscure the position estimation. $\mathcal{O}_{\text{BE}}^{\mathbf{x}_t}$ represents the integrated positioning information actually essential for target localization, aggregated from the measurements of all position-inaccurate PDs. It quantifies the system's ability to precisely distinguish positional differences within the VLP framework. Essentially, $\mathcal{O}_{\text{BE}}^{\mathbf{x}_t}$ can be understood as the refined outcome of raw measurement data, initially acquired without considering the PD position errors, and subsequently corrected using the prior accuracy of the PD positions, denoted by $\mathbf{U}_t^i$. The corrective influence of $\mathbf{U}_t^i$ on $\mathcal{O}_{\text{BE}}^{\mathbf{x}_t}$ is evident in (33) and (101). A zero-valued $\mathcal{O}_{\text{BE}}^{\mathbf{x}_t}$ implies the unobservability of $\mathbf{x}_t$, whereas a good performance is associated with a large $\mathcal{O}_{\text{BE}}^{\mathbf{x}_t}$.

Remark 3: From (19)–(37), it can be seen that the number of obscured PDs within the influenced range by the reference LED position, i.e., Ω_t, plays a pivotal role in determining $\mathcal{O}_{\text{BE}}^{\mathbf{x}_t}$. Thus, a judicious selection of reference LED and an efficient deployment of PDs are essential for achieving the performance equilibrium, particularly in resource-constrained scenarios.

B. Sensor Calibration Analysis

1) Closed-Form CRLB: The calibration error $||\hat{\mathbf{r}}_t^i - \mathbf{r}_t^i||_2$ is bounded by $\mathbb{E}_{\mathbf{f}_t}\{||\hat{\mathbf{r}}_t^i - \mathbf{r}_t^i||_2\} \geqslant \text{tr}(\mathcal{B}_{\text{BE}}^{\mathbf{r}_t^i})^{\frac{1}{2}}$, in which $\mathcal{B}_{\text{BE}}^{\mathbf{r}_t^i} \in \mathbb{S}^3$ denoting the closed-form BE-based CRLB on the PD position calibration error can be given by Theorem 2.

Theorem 2 (Single-Time-Slot Calibration CRLB): At time t, the CRLB $\mathcal{B}_{\text{BE}}^{\mathbf{r}_t^i}$ of the blocked PD position follows as

$$\mathcal{B}_{\text{BE}}^{\mathbf{r}_t^i} = \left(\varpi \cdot \mathcal{Q}_{\text{BE}}^{\mathbf{r}_t^i,\mathbf{t}_t} + \vartheta \cdot \mathcal{Q}_{\text{BE}}^{\mathbf{r}_t^i,\mathbf{x}_t} + \mathbf{U}_t^i\right)^{-1}, \tag{38}$$

where the SNR $\varpi = \Gamma\gamma\Psi_R^2 \in \mathbb{R}$ is defined unlike ϑ in (13), while the corresponding equivalent measurement information, denoted as $\mathcal{Q}_{\text{BE}}^{\mathbf{r}_t^i,\mathbf{t}_t} \in \mathbb{S}^3$ and $\mathcal{Q}_{\text{BE}}^{\mathbf{r}_t^i,\mathbf{x}_t} \in \mathbb{S}^3$, can be derived from (39) and (40), respectively. Two components of observational data, sourced from the LED collection Φ_t^i and the target $\mathbf{x}_t$, can be defined as $\mathcal{R}_{\text{obs}}^{\mathbf{t}_t} = \varpi \cdot \mathcal{Q}_{\text{BE}}^{\mathbf{r}_t^i,\mathbf{t}_t} \in \mathbb{S}^3$ and $\mathcal{R}_{\text{obs}}^{\mathbf{x}_t} = \vartheta \cdot \mathcal{Q}_{\text{BE}}^{\mathbf{r}_t^i,\mathbf{x}_t} \in \mathbb{S}^3$, respectively. Accordingly, the overall observation information related to the PD position $\mathbf{r}_t^i$ can be denoted as a summation form, i.e., $\mathcal{R}_{\text{obs}}^{\mathbf{r}_t^i} = \mathcal{R}_{\text{obs}}^{\mathbf{t}_t} + \mathcal{R}_{\text{obs}}^{\mathbf{x}_t} \in \mathbb{S}^3$.

Proof: Its derivation is similar to that given in Appendix A. Indeed, $\mathcal{Q}_{\text{BE}}^{\mathbf{r}_t^i,\mathbf{t}_t}$ and $\mathcal{Q}_{\text{BE}}^{\mathbf{r}_t^i,\mathbf{x}_t}$ in Theorem 2 are formed as

$$\begin{aligned}\mathcal{Q}_{\text{BE}}^{\mathbf{r}_t^i,\mathbf{t}_t} &= \mathbf{D}(\mathbf{r}_t^i)\mathbf{K}^\circ(\mathbf{u})\mathbf{K}^\circ(\mathbf{u})^\top\mathbf{D}(\mathbf{r}_t^i)^\top \\ &\quad - \mathbf{P}(\mathbf{r}_t^i,\mathbf{t}_t)\mathbf{W}(\mathbf{t}_t)\mathbf{P}(\mathbf{r}_t^i,\mathbf{t}_t)^\top,\end{aligned} \tag{39}$$

$$\mathcal{Q}_{\text{BE}}^{\mathbf{r}_t^i,\mathbf{x}_t} = \mathbf{q}(\mathbf{r}_t^i)\mathbf{u}^i\text{o}(\mathbf{x}_t)\mathbf{u}^{i\top}\mathbf{q}(\mathbf{r}_t^i)^\top, \tag{40}$$

where $\mathbf{D}(\mathbf{r}_t^i)$ and $\mathbf{K}^\circ(\mathbf{u})$ are given, respectively, by

$$\mathbf{D}(\mathbf{r}_t^i) = \left[(\rho_t^{i,j})^{-3}\mathbf{d}_t^{j,i}(\mathbf{r}_t^i) \mid \forall j \in \Phi_t^i\right] \in \mathbb{R}^{3\times 3\Phi_t}, \tag{41}$$

$$\begin{aligned}\mathbf{d}_t^{j,i}(\mathbf{r}_t^i) \in \mathbb{R}^{3\times 3} &= d(d+1)(\mathbf{e}_{\mathbf{r},t}^{i,j\top}\mathbf{v}^j)^{d-1}\mathbf{v}^j\mathbf{e}_{\mathbf{r},t}^{i,j\top} \\ &\quad + (d+1)(\mathbf{e}_{\mathbf{r},t}^{i,j\top}\mathbf{v}^j)^d\mathbf{I}_3 \\ &\quad - (d+3)(d+1)(\mathbf{e}_{\mathbf{r},t}^{i,j\top}\mathbf{v}^j)^d\mathbf{e}_{\mathbf{r},t}^{i,j}\mathbf{e}_{\mathbf{r},t}^{i,j\top},\end{aligned} \tag{42}$$

$$\mathbf{K}^\circ(\mathbf{u}) = \mathbf{I}_{\Phi_t} \otimes \mathbf{u}^i \in \mathbb{R}^{3\Phi_t\times\Phi_t}, \tag{43}$$

while $\mathbf{P}(\mathbf{r}_t^i,\mathbf{t}_t)$, $\mathbf{W}(\mathbf{t}_t)$ and $\text{o}(\mathbf{x}_t)$ are defined, respectively, as

$$\mathbf{P}(\mathbf{r}_t^i,\mathbf{t}_t) = \left[\mathbf{d}(\mathbf{r}_t^i)\mathbf{u}^i\mathbf{u}^{i\top}\mathbf{c}(\mathbf{t}_t^j)^\top \mid \forall j \in \Phi_t^i\right] \in \mathbb{R}^{3\times 3\Phi_t}, \tag{44}$$

$$\mathbf{d}(\mathbf{r}_t^i) = (\rho_t^{i,j})^{-3}\mathbf{d}_t^{j,i}(\mathbf{r}_t^i) \in \mathbb{R}^{3\times 3}, \tag{45}$$

$$\mathbf{c}(\mathbf{t}_t^j) = -(\rho_t^{i,j})^{-3}\mathbf{d}_t^{j,i}(\mathbf{r}_t^i) = -\mathbf{d}(\mathbf{r}_t^i) \in \mathbb{R}^{3\times 3}, \tag{46}$$

$$\mathbf{W}(\mathbf{t}_t) = diag\big(\mathbf{w}(\mathbf{t}_t^1), \cdots, \mathbf{w}(\mathbf{t}_t^{\Phi_t})\big) \in \mathbb{R}^{3\Phi_t\times 3\Phi_t}, \tag{47}$$

$$\mathbf{w}(\mathbf{t}_t^j) = \left(\mathbf{c}(\mathbf{t}_t^j)\mathbf{u}^i\mathbf{u}^{i\top}\mathbf{c}(\mathbf{t}_t^j)^\top\right)^{-1} \in \mathbb{R}^{3\times 3}, \tag{48}$$

$$\text{o}(\mathbf{x}_t) = 1 - \mathbf{u}^{i\top}\mathbf{f}(\mathbf{x}_t)^\top\mathbf{O}(\mathbf{x}_t)\mathbf{f}(\mathbf{x}_t)\mathbf{u}^i \in \mathbb{R}, \tag{49}$$

$$\mathbf{O}(\mathbf{x}_t) = \left(\mathbf{f}(\mathbf{x}_t)\mathbf{u}^i\mathbf{u}^{i\top}\mathbf{f}(\mathbf{x}_t)^\top + \vartheta^{-1}\mathcal{H}_{\text{BE}}^{\mathbf{x}_t}\right)^{-1} \in \mathbb{R}^{3\times 3}, \tag{50}$$

where $\mathbf{f}(\mathbf{x}_t)$ and $\mathcal{H}_{\text{BE}}^{\mathbf{x}_t}$ are given by (25) and (13), respectively.

2) Comments on Error Limit: From Theorem 2, the associated remarks on the error bounds for PD position calibration at each moment can be derived as follows.

Remark 4: Based on (38), it becomes apparent that $\mathcal{B}_{\text{BE}}^{\mathbf{r}_t^i} = (\mathcal{H}_{\text{BE}}^{\mathbf{r}_t^i})^{-1}$, in which $\mathcal{H}_{\text{BE}}^{\mathbf{r}_t^i} = \mathcal{R}_{\text{obs}}^{\mathbf{t}_t} + \mathcal{R}_{\text{obs}}^{\mathbf{x}_t} + \mathbf{U}_t^i$. Thus, (38) elucidates the principle of error evolution regarding each PD position calibration, denoted by the transition from $(\mathbf{U}_t^i)^{-1}$ to $\mathcal{B}_{\text{BE}}^{\mathbf{r}_t^i}$. Compared to (13) in Theorem 1, which emphasizes the convergence analysis, (38) establishes part of the theoretical groundwork for the asymptotic analysis shown in Section V.

Remark 5: From (38), it is evident that the current calibration precision $\mathcal{H}_{\text{BE}}^{\mathbf{r}_t^i}$ is primarily influenced by three factors: (i) the observation information $\mathcal{R}_{\text{obs}}^{\mathbf{t}_t}$ from the Φ_t eligible LEDs, represented as $\varpi \cdot \mathcal{Q}_{\text{BE}}^{\mathbf{r}_t^i,\mathbf{t}_t}$; (ii) the observation information $\mathcal{R}_{\text{obs}}^{\mathbf{x}_t}$ from the moving target, denoted by $\vartheta \cdot \mathcal{Q}_{\text{BE}}^{\mathbf{r}_t^i,\mathbf{x}_t}$; and (iii) the state accuracy $\mathbf{U}_t^i$. Additionally, the PD location error is reflected only in $\mathcal{R}_{\text{obs}}^{\mathbf{x}_t}$, as all LED positions are assumed to be accurate.

IV. Convergence Analysis of Error Evolution

This section helps to reveal the long-term VLP performance by elucidating why the VLP error converges, how it achieves convergence, and the properties of its converged state.

A. Convergence Establishment

Owing to target mobility characterized by χ, the prediction information $\mathcal{I}_{\text{pre}}^{\mathbf{x}_t}$ must be less than the most recent posterior $\mathcal{H}_{\text{BE}}^{\mathbf{x}_{t-1}}$, indicating an information loss during the prediction phase, which can be quantified by $\mathcal{J}_{\mathbf{x}_t} \in \mathbb{S}^3$ in Corollary 1.

Corollary 1 (Non-Negative Loss): $\mathcal{J}_{\mathbf{x}_t} \succeq \mathbf{0}$ is given by

$$\mathcal{J}_{\mathbf{x}_t} = \mathcal{H}_{\text{BE}}^{\mathbf{x}_{t-1}} - \mathcal{I}_{\text{pre}}^{\mathbf{x}_t} = (\mathcal{B}_{\text{BE}}^{\mathbf{x}_{t-1}}\chi\mathcal{B}_{\text{BE}}^{\mathbf{x}_{t-1}} + \mathcal{B}_{\text{BE}}^{\mathbf{x}_{t-1}})^{-1}. \tag{51}$$

Proof: (51) is derived based on the matrix inversion lemma, i.e., $\mathcal{I}_{\text{pre}}^{\mathbf{x}_t} = (\mathcal{B}_{\text{BE}}^{\mathbf{x}_{t-1}})^{-1} - (\mathcal{B}_{\text{BE}}^{\mathbf{x}_{t-1}}\chi\mathcal{B}_{\text{BE}}^{\mathbf{x}_{t-1}} + \mathcal{B}_{\text{BE}}^{\mathbf{x}_{t-1}})^{-1}$.

Theorem 3 (Monotonically Non-Increasing CRLB): If the observation information $\mathcal{L}_{\text{obs}}^{\mathbf{x}_t}$ is not less than the loss $\mathcal{J}_{\mathbf{x}_t}$ in (51), i.e., $\mathcal{L}_{\text{obs}}^{\mathbf{x}_t} \succeq \mathcal{J}_{\mathbf{x}_t}$, the error tends to decrease over time,

$$\mathcal{B}_{\text{BE}}^{\mathbf{x}_t} \preceq \mathcal{B}_{\text{BE}}^{\mathbf{x}_{t-1}}, \ \forall t > 0, \ \text{given } \mathcal{L}_{\text{obs}}^{\mathbf{x}_t} \succeq \mathcal{J}_{\mathbf{x}_t}. \quad (52)$$

Proof: Refer to Appendix B.

Remark 6: To ensure the convergence of $\mathcal{B}_{\text{BE}}^{\mathbf{x}_t}$, it is imperative that $\mathcal{L}_{\text{obs}}^{\mathbf{x}_t}$ compensates for $\mathcal{J}_{\mathbf{x}_t}$. As per (51), $\mathcal{J}_{\mathbf{x}_t}$ relies on the previous $\mathcal{B}_{\text{BE}}^{\mathbf{x}_{t-1}}$ and the inherent target shift accuracy $\boldsymbol{\chi}$, while (13) indicates that $\mathcal{L}_{\text{obs}}^{\mathbf{x}_t}$ comprises solely external measurement information related to the PD set Ω_t^j. This means proper PD deployment is essential to gain ideal performance.

Theorem 4 (Bounded CRLB): If $\mathcal{L}_{\text{obs}}^{\mathbf{x}_t} \to \mathcal{L}_{\text{obs}}^{\mathbf{x}*}$, as $t \to \infty$ is satisfied, where $\mathcal{L}_{\text{obs}}^{\mathbf{x}*} \in \mathbb{S}_+^3$ is its limit, $\mathcal{B}_{\text{BE}}^{\mathbf{x}_t}$ converges to the stable $\mathcal{B}_{\text{BE}}^{\mathbf{x}*} \in \mathbb{S}^3$ at the linear convergence rate $\eta_{\mathbf{x}} \in (0,1)$ as

$$\eta_{\mathbf{x}} = \lim_{t\to\infty} \frac{||\mathcal{B}_{\text{BE}}^{\mathbf{x}_t} - \mathcal{B}_{\text{BE}}^{\mathbf{x}_{t+1}}||_F}{||\mathcal{B}_{\text{BE}}^{\mathbf{x}_{t-1}} - \mathcal{B}_{\text{BE}}^{\mathbf{x}_t}||_F} = 1 - \zeta_{\mathbf{x}}, \quad (53)$$

where (53) is essentially derived by substituting $\mathcal{B}_{\text{BE}}^{\mathbf{x}*}$ into the quotient convergence definition [43], and $||\bullet||_F$ is Frobenius norm, while $\zeta_{\mathbf{x}} \in (0,1)$ is a constant defined as follows, i.e.,

$$\zeta_{\mathbf{x}} = \left|\left|\mathcal{A}_{\text{BE}}^{\mathbf{x}*}\left(\mathcal{A}_{\text{BE}}^{\mathbf{x}*}+(\mathcal{L}_{\text{obs}}^{\mathbf{x}*})^{-1}\right)^{-2}\left(\mathcal{A}_{\text{BE}}^{\mathbf{x}*}+2(\mathcal{L}_{\text{obs}}^{\mathbf{x}*})^{-1}\right)\right|\right|_F, \quad (54)$$

where $\mathcal{A}_{\text{BE}}^{\mathbf{x}*} \in \mathbb{S}^3$ is denoted as $\mathcal{A}_{\text{BE}}^{\mathbf{x}*} = \mathcal{B}_{\text{BE}}^{\mathbf{x}*} + \boldsymbol{\chi}^{-1}$. From Theorem 3 and 4, $\mathcal{B}_{\text{BE}}^{\mathbf{x}_t}$ converges by monotonically bounded.

Proof: See Appendix C.

Remark 7: It is evident that $\mathcal{B}_{\text{BE}}^{\mathbf{x}_t}$ will gradually diminish its decrease rate over time, ultimately converging to a stable limit, which symbolizes the long-term performance. Regarding PD calibration, as per (38) , given both $\mathcal{Q}_{\text{BE}}^{\mathbf{r}_t^i,\mathbf{t}_t} \succeq \mathbf{0}$ and $\mathcal{Q}_{\text{BE}}^{\mathbf{r}_t^i,\mathbf{x}_t} \succeq \mathbf{0}$, it is imperative to ensure that both $\boldsymbol{\Gamma}\gamma > 0$ and $\boldsymbol{\Lambda}\lambda > 0$. This requirement helps satisfy the condition $(\mathcal{B}_{\text{BE}}^{\mathbf{r}_t^i})^{-1} \succ \mathbf{U}_t^i$, thus enabling calibration errors to converge. As a result, the inaccuracies in the PDs are progressively refined, leading to increased precision over time.

B. Convergence State

The information loss is time-varying by the corollary below.

Corollary 2 (Increasing Information Loss): For the decreasing $\mathcal{B}_{\text{BE}}^{\mathbf{x}_t}$ in (52), the information loss $\mathcal{J}_{\mathbf{x}_t}$ in (51) follows as

$$\mathcal{J}_{\mathbf{x}_t} \succeq \mathcal{J}_{\mathbf{x}_{t-1}}, \ \text{given } \mathcal{B}_{\text{BE}}^{\mathbf{x}_t} \preceq \mathcal{B}_{\text{BE}}^{\mathbf{x}_{t-1}}. \quad (55)$$

Proof: This can be proved by $\mathcal{J}_{\mathbf{x}_t} \propto (\mathcal{B}_{\text{BE}}^{\mathbf{x}_{t-1}})^{-1}$ from (51).

Remark 8: As $\mathcal{H}_{\text{BE}}^{\mathbf{x}_{t-1}} \propto (\mathcal{B}_{\text{BE}}^{\mathbf{x}_{t-1}})^{-1}$ and $\mathcal{J}_{\mathbf{x}_t} \propto (\mathcal{B}_{\text{BE}}^{\mathbf{x}_{t-1}})^{-1}$, we have $\mathcal{J}_{\mathbf{x}_t} \propto \mathcal{H}_{\text{BE}}^{\mathbf{x}_{t-1}}$. Namely, the information loss $\mathcal{J}_{\mathbf{x}_t}$ increases as the VLP accuracy $\mathcal{H}_{\text{BE}}^{\mathbf{x}_{t-1}}$ improves over time until it equals the observation information $\mathcal{L}_{\text{obs}}^{\mathbf{x}_t}$. Originally, $\mathcal{J}_{\mathbf{x}_t}$ is significantly less than $\mathcal{L}_{\text{obs}}^{\mathbf{x}_t}$ during VLP, resulting in an obvious convergence trend. However, as $\mathcal{J}_{\mathbf{x}_t}$ gradually increases, the rate of decline slows down. If $\mathcal{J}_{\mathbf{x}_t}$ surpasses $\mathcal{L}_{\text{obs}}^{\mathbf{x}_t}$, failing to meet the condition of Theorem 3, the immediate value $\mathcal{H}_{\text{BE}}^{\mathbf{x}_{t-1}}$ slightly drops. Consequently, the corresponding $\mathcal{J}_{\mathbf{x}_t}$ decreases until $\mathcal{L}_{\text{obs}}^{\mathbf{x}_t}$ compensates for it, allowing continued convergence. Therefore, $\mathcal{L}_{\text{obs}}^{\mathbf{x}_t}$ and $\mathcal{J}_{\mathbf{x}_t}$ exhibit a dynamic and time-varying complementarity, resulting in a fluctuating stabilization of the VLP error $\mathcal{B}_{\text{BE}}^{\mathbf{x}_t}$. This behavior indicates that the VLP system is capable of accommodating sudden localization failures while autonomously guiding performance toward a stable balance.

Theorem 5 (Closed-Form Balance State): The VLP error $\mathcal{B}_{\text{BE}}^{\mathbf{x}_t}$ will converge over time to a stable state $\mathcal{B}_{\text{BE}}^{\mathbf{x}*}$ defined as

$$\mathcal{B}_{\text{BE}}^{\mathbf{x}*} = \frac{1}{2}\boldsymbol{\chi}^{-\frac{1}{2}}\left(\mathbf{I}_3 + 4\boldsymbol{\chi}^{\frac{1}{2}}(\mathcal{L}_{\text{obs}}^{\mathbf{x}_t})^{-1}\boldsymbol{\chi}^{\frac{1}{2}}\right)^{\frac{1}{2}}\boldsymbol{\chi}^{-\frac{1}{2}} - \frac{1}{2}\boldsymbol{\chi}^{-1}, \quad (56)$$

where $\boldsymbol{\chi}$ and $\mathcal{L}_{\text{obs}}^{\mathbf{x}_t}$ given by (13) are assumed in $\mathbb{S}_+^3$.

Proof: See the proof in Appendix D.

Remark 9: The above error stable state $\mathcal{B}_{\text{BE}}^{\mathbf{x}*}$ is completely determined by the transition precision $\boldsymbol{\chi}$ and the observation information $\mathcal{L}_{\text{obs}}^{\mathbf{x}_t}$. From (56), it can be inferred that $\mathcal{B}_{\text{BE}}^{\mathbf{x}*}$ is a Lipschiz-continuous function with respect to $\mathcal{L}_{\text{obs}}^{\mathbf{x}_t}$. In practical scenarios, where the matrices in (56) are full rank, the bounded $\mathcal{B}_{\text{BE}}^{\mathbf{x}*}$ is entirely contingent upon the boundedness of $\mathcal{L}_{\text{obs}}^{\mathbf{x}_t}$.

Theorem 6 (Bounded Balance State): $\mathcal{B}_{\text{BE}}^{\mathbf{x}*}$ can bound as

$$(\mathcal{L}_{\text{obs}}^{\mathbf{x},\min} + \boldsymbol{\chi})^{-1} \preceq \mathcal{B}_{\text{BE}}^{\mathbf{x}*} \preceq (\mathcal{L}_{\text{obs}}^{\mathbf{x},\max})^{-1}, \quad (57)$$

where the leftmost and rightmost sides of it respectively define the lower and upper bounds, i.e., $\mathcal{B}_{\text{BE}}^{\mathbf{x}*,\min}$ and $\mathcal{B}_{\text{BE}}^{\mathbf{x}*,\max}$, while $\mathcal{L}_{\text{obs}}^{\mathbf{x},\min}$ and $\mathcal{L}_{\text{obs}}^{\mathbf{x},\max}$ are the minimum and maximum of $\mathcal{L}_{\text{obs}}^{\mathbf{x}_t}$.

Proof: Refer to Appendix E.

Remark 10: Since the observation information $\mathcal{L}_{\text{obs}}^{\mathbf{x}_t}$ typically varies within a bounded area, the fluctuation range of the error stable state $\mathcal{B}_{\text{BE}}^{\mathbf{x}*}$ is directly influenced by the variation in $\mathcal{L}_{\text{obs}}^{\mathbf{x}_t}$.

C. Convergence Properties

We focus on the convergence rate and the $\mathcal{B}_{\text{BE}}^{\mathbf{x}*}$ uncertainty.

Corollary 3 (Convergence Rate Characteristics): The rate $\eta_{\mathbf{x}}$ given by (53) is proportional to $\boldsymbol{\chi}$ and $\mathcal{L}_{\text{obs}}^{\mathbf{x}_t}$ as follows,

$$\eta_{\mathbf{x}} \propto \mathcal{L}_{\text{obs}}^{\mathbf{x}_t}, \ \eta_{\mathbf{x}} \propto \boldsymbol{\chi}. \quad (58)$$

Proof: By (13) and (56), we get $\mathcal{B}_{\text{BE}}^{\mathbf{x}_t}, \mathcal{B}_{\text{BE}}^{\mathbf{x}*} \propto (\mathcal{L}_{\text{obs}}^{\mathbf{x}_t})^{-1}$ and $\mathcal{B}_{\text{BE}}^{\mathbf{x}_t}, \mathcal{B}_{\text{BE}}^{\mathbf{x}*} \propto \boldsymbol{\chi}^{-1}$. Then from (54), we can have $\zeta_{\mathbf{x}} \propto \mathcal{A}_{\text{BE}}^{\mathbf{x}*}$, $\mathcal{A}_{\text{BE}}^{\mathbf{x}*} \propto (\mathcal{L}_{\text{obs}}^{\mathbf{x}_t})^{-1}$ and $\mathcal{A}_{\text{BE}}^{\mathbf{x}*} \propto \boldsymbol{\chi}^{-1}$. As $\eta_{\mathbf{x}} \propto (\zeta_{\mathbf{x}})^{-1}$, it is hold.

Remark 11: Larger values of $\mathcal{L}_{\text{obs}}^{\mathbf{x}_t}$ and $\boldsymbol{\chi}$ mean slower final error convergence and a lower overall error $(\mathcal{B}_{\text{BE}}^{\mathbf{x}_t}, \mathcal{B}_{\text{BE}}^{\mathbf{x}*})$. That is, even though the convergence rate per unit time is higher, the time required to achieve steady-state convergence is longer.

Let $\mathbf{B}_t = \text{cov}(\mathcal{B}_{\text{BE}}^{\mathbf{x}*}) \in \mathbb{S}^3$ be the fluctuation degree of the stable $\mathcal{B}_{\text{BE}}^{\mathbf{x}*}$, where $\text{cov}(\bullet)$ is the covariance matrix. Then, the following corollary indicates the role of influences on $\mathbf{B}_t$.

Corollary 4 (Stable State Volatility): $\mathbf{B}_t$ scales with them as

$$\mathbf{B}_t \propto (\mathcal{L}_{\text{obs}}^{\mathbf{x}_t})^{-1}, \ \mathbf{B}_t \propto \boldsymbol{\chi}^{-1}. \quad (59)$$

Proof: From Remark 11, it can be obtained since $\mathbf{B}_t \propto \mathcal{B}_{\text{BE}}^{\mathbf{x}*}$.

Remark 12: Larger values of $\mathcal{L}_{\text{obs}}^{\mathbf{x}_t}$ and $\boldsymbol{\chi}$ lead to a smaller $\mathbf{B}_t$, meaning that the error stable state $\mathcal{B}_{\text{BE}}^{\mathbf{x}*}$ becomes more steady as $\mathcal{L}_{\text{obs}}^{\mathbf{x}_t}$ and $\boldsymbol{\chi}$ increase.

V. Asymptotic Analysis of Error Evolution

In this section, we conduct a systematic asymptotic analysis to evaluate the influence of several crucial system parameters (e.g., $\boldsymbol{\vartheta}; \boldsymbol{\varpi}$, $\rho_t^{i,j}$, $\Omega_t; \Phi_t$, and $\mathbf{U}_t^i$) on VLP performance.

A. Asymptotic Limits of Localization and Calibration CRLB

Based on (13) and (38), it is evident that both localization and calibration accuracies are affected by the respective noise factors, represented by $\boldsymbol{\vartheta}$ and $\boldsymbol{\varpi}$. Hence, the corollary below elucidates the impact of noise precision on VLP performance.

Corollary 5 (The Effect of Noise Precision): The VLP error on target localization and PD position calibration follows as

$$\lim_{\boldsymbol{\vartheta}\to 0}\boldsymbol{\mathcal{B}}_{\mathrm{BE}}^{\mathbf{x}_t}=\boldsymbol{\mathcal{B}}_{\mathrm{BE}}^{\mathbf{x}_{t-1}}+\boldsymbol{\chi}^{-1},\ \lim_{\boldsymbol{\vartheta}\to \infty}\boldsymbol{\mathcal{B}}_{\mathrm{BE}}^{\mathbf{x}_t}=(\boldsymbol{\mathcal{L}}_{\mathrm{obs}}^{\mathbf{x}_t}+\boldsymbol{\chi})^{-1},\quad (60)$$

$$\lim_{\substack{\boldsymbol{\varpi}\to 0,\\ \boldsymbol{\vartheta}\to 0}}\boldsymbol{\mathcal{B}}_{\mathrm{BE}}^{\mathbf{r}_t^i}=(\mathbf{U}_t^i)^{-1},\ \lim_{\substack{\boldsymbol{\varpi}\to \infty,\\ \boldsymbol{\vartheta}\to \infty}}\boldsymbol{\mathcal{B}}_{\mathrm{BE}}^{\mathbf{r}_t^i}=\mathbf{0},\quad (61)$$

$$\mathrm{tr}(\boldsymbol{\mathcal{B}}_{\mathrm{BE}}^{\mathbf{x}_t})\propto\boldsymbol{\vartheta}^{-1},\ \mathrm{tr}(\boldsymbol{\mathcal{B}}_{\mathrm{BE}}^{\mathbf{r}_t^i})\propto\boldsymbol{\varpi}^{-1}.\quad (62)$$

Proof: This directly follows from Theorem 1 and 2.

Remark 13: The VLP error decreases as the noise accuracy increases until it reaches a limit defined by the inverse of accuracy itself. In cases of complete observation failure, localization information relies solely on predicted data. Conversely, when observations are precise and sufficient, any loss of information due to environmental disturbances can be disregarded.

Let $\rho_{t,\min}^{i,j}=\min\{\rho_t^{i,j}\,|\,\forall i\in\Omega_t\}$ be the minimum distance of $\rho_t^{i,j}$ during localization and denote $\rho_{t,\min}^{j,i}=\min\{\rho_t^{i,j}\,|\,\forall j\in\Phi_t\}$ as the minimum distance of $\rho_t^{i,j}$ during calibration, then the following corollary can show the error scaling rule w.r.t. $\rho_t^{i,j}$.

Corollary 6 (The Effect of Transmission Distance): As the minimum $\rho_{t,\min}^{i,j}\to\infty$ and $\rho_{t,\min}^{j,i}\to\infty$, the error follows as

$$\mathrm{tr}(\boldsymbol{\mathcal{B}}_{\mathrm{BE}}^{\mathbf{x}_t})\propto(\rho_{t,\max}^{i,j})^4,\ \mathrm{tr}(\boldsymbol{\mathcal{B}}_{\mathrm{BE}}^{\mathbf{r}_t^i})\propto(\rho_{t,\max}^{j,i})^6,\quad (63)$$

where $\rho_{t,\max}^{i,j}=\max\{\rho_t^{i,j}|\forall i\in\Omega_t\},\rho_{t,\max}^{j,i}=\max\{\rho_t^{i,j}|\forall j\in\Phi_t\}$.

Proof: See Appendix F.

Remark 14: This indicates that the target localization error is increasing with the transmission distance in the fourth power, while the sensor calibration error is increasing with its sixth power, which essentially rests with the intrinsic physical model of signal propagation. Corollary 6 also suggests that a small monitored area is preferable and establishes guidelines for assessing rational deployment principles within a defined area.

Regarding the error scaling rule w.r.t. reference set size (Ω_t, Φ_t), the below corollary clearly indicates its relationship.

Corollary 7 (The Effect of Reference Set Size): As all LEDs and PDs are uniformly deployed, $\bar{\boldsymbol{\mathcal{H}}}_{\mathrm{BE}}^{\mathbf{x}_t}, \bar{\boldsymbol{\mathcal{H}}}_{\mathrm{BE}}^{\mathbf{r}_t^i}\in\mathbb{S}^3$ meaning the precision scale linearly with the reference set sizes, namely,

$$\bar{\boldsymbol{\mathcal{H}}}_{\mathrm{BE}}^{\mathbf{x}_t}=\Omega_t\bar{\boldsymbol{\mathcal{L}}}_{\mathrm{obs}}^{\mathbf{x}_t}+\boldsymbol{\mathcal{I}}_{\mathrm{pre}}^{\mathbf{x}_t},\ \bar{\boldsymbol{\mathcal{H}}}_{\mathrm{BE}}^{\mathbf{r}_t^i}=\Phi_t\bar{\boldsymbol{\mathcal{R}}}_{\mathrm{obs}}^{\mathbf{t}_t}+\boldsymbol{\mathcal{R}}_{\mathrm{obs}}^{\mathbf{x}_t}+\mathbf{U}_t^i,\quad (64)$$

where $\bar{\boldsymbol{\mathcal{L}}}_{\mathrm{obs}}^{\mathbf{x}_t},\bar{\boldsymbol{\mathcal{R}}}_{\mathrm{obs}}^{\mathbf{t}_t}\in\mathbb{S}^3$ are the relevant assumed long-term averaged equivalent dominant observation information expectation for localization and calibration. Then, the error obeys

$$\mathrm{tr}(\boldsymbol{\mathcal{B}}_{\mathrm{BE}}^{\mathbf{x}_t})\propto\Omega_t^{-1},\ \mathrm{tr}(\boldsymbol{\mathcal{B}}_{\mathrm{BE}}^{\mathbf{r}_t^i})\propto\Phi_t^{-1}.\quad (65)$$

Proof: See the proof in Appendix G.

Remark 15: This reveals a performance benchmark regarding the number of nodes deployed, i.e., the VLP error bounds will decrease at a linear rate as the reference set size increases. That is, even if the PD position is inaccurate, the addition of this positioning PD still improves the localization accuracy, i.e., an additional reference node represents a performance gain ($\bar{\boldsymbol{\mathcal{L}}}_{\mathrm{obs}}^{\mathbf{x}_t}$, $\bar{\boldsymbol{\mathcal{R}}}_{\mathrm{obs}}^{\mathbf{t}_t}$). Even if all PD positions are inaccurate, the errors can be arbitrarily small as long as the reference set size Ω_t is large enough. Besides, the performance gain from an increased reference set size will be progressively negligible, i.e., factors such as noise accuracy will become the primary error sources if the observation information paths are sufficient.

Considering the PD position errors, we propose the following corollary regarding the impact of $\mathbf{U}_t^i$ on VLP performance. It is important to re-emphasize that a larger value of $\mathbf{U}_t^i$ means a more accurate PD position and a smaller impact on VLP.

Corollary 8 (The Effect of Location Uncertainties): As the position precision $\mathbf{U}_t^i$ tends to $\mathbf{0}$ and $\boldsymbol{\infty}$, the error follows that

$$\lim_{\substack{\mathbf{U}_t^i\to\mathbf{0},\\ \forall i\in\Omega_t}}\boldsymbol{\mathcal{B}}_{\mathrm{BE}}^{\mathbf{x}_t}=(\boldsymbol{\mathcal{I}}_{\mathrm{pre}}^{\mathbf{x}_t})^{-1},\ \lim_{\substack{\mathbf{U}_t^i\to\boldsymbol{\infty},\\ \forall i\in\Omega_t}}\boldsymbol{\mathcal{B}}_{\mathrm{BE}}^{\mathbf{x}_t}=(\boldsymbol{\mathcal{H}}_{\boldsymbol{\infty}}^{\mathbf{x}_t})^{-1}=\boldsymbol{\mathcal{B}}_{\boldsymbol{\infty}}^{\mathbf{x}_t},\quad (66)$$

$$\lim_{\substack{\mathbf{U}_t^i\to\mathbf{0},\\ \forall i\in\Omega_t}}\boldsymbol{\mathcal{B}}_{\mathrm{BE}}^{\mathbf{r}_t^i}=(\boldsymbol{\mathcal{R}}_{\mathrm{obs}}^{\mathbf{t}_t}+\boldsymbol{\mathcal{R}}_{\mathrm{obs}}^{\mathbf{x}_t})^{-1},\ \lim_{\substack{\mathbf{U}_t^i\to\boldsymbol{\infty},\\ \forall i\in\Omega_t}}\boldsymbol{\mathcal{B}}_{\mathrm{BE}}^{\mathbf{r}_t^i}=\mathbf{0},\quad (67)$$

$$\mathrm{tr}(\boldsymbol{\mathcal{B}}_{\mathrm{BE}}^{\mathbf{x}_t})\propto(\mathbf{U}_t^i)^{-1},\ \mathrm{tr}(\boldsymbol{\mathcal{B}}_{\mathrm{BE}}^{\mathbf{r}_t^i})\propto(\mathbf{U}_t^i)^{-1},\quad (68)$$

where $\boldsymbol{\mathcal{H}}_{\boldsymbol{\infty}}^{\mathbf{x}_t}\in\mathbb{S}^3$ is defined as $\boldsymbol{\mathcal{H}}_{\boldsymbol{\infty}}^{\mathbf{x}_t}=\mathbf{V}(\mathbf{x}_t)+\boldsymbol{\mathcal{I}}_{\mathrm{pre}}^{\mathbf{x}_t}$ and then $\boldsymbol{\mathcal{B}}_{\boldsymbol{\infty}}^{\mathbf{x}_t}\in\mathbb{S}^3$ is attained, in which $\mathbf{V}(\mathbf{x}_t)\in\mathbb{S}^3$ can be given by

$$\mathbf{V}(\mathbf{x}_t)=\boldsymbol{\vartheta}\cdot\mathbf{F}(\mathbf{x}_t)\mathbf{K}(\mathbf{u})\mathbf{S}(\mathbf{t}^j)\mathbf{K}(\mathbf{u})^\top\mathbf{F}(\mathbf{x}_t)^\top,\quad (69)$$

$$\mathbf{S}(\mathbf{t}^j)=\mathbf{I}_{\Omega_t}-\mathbf{K}(\mathbf{u})^\top\mathbf{J}(\mathbf{t}^j)^\top\mathbf{T}(\mathbf{t}^j)\mathbf{J}(\mathbf{t}^j)\mathbf{K}(\mathbf{u})\in\mathbb{R}^{\Omega_t\times\Omega_t},\quad (70)$$

$$\mathbf{T}(\mathbf{t}^j)=\left(\mathbf{J}(\mathbf{t}^j)\mathbf{K}(\mathbf{u})\mathbf{K}(\mathbf{u})^\top\mathbf{J}(\mathbf{t}^j)^\top\right)^{-1}\in\mathbb{R}^{3\times 3},\quad (71)$$

where $\mathbf{F}(\mathbf{x}_t)$, $\mathbf{K}(\mathbf{u})$ and $\mathbf{J}(\mathbf{t}^j)$ are denoted in Theorem 1.

Proof: Refer to Appendix H.

Remark 16: This implies that the VLP error increases with the PD position error. The asymptotic limit $\boldsymbol{\mathcal{B}}_{\boldsymbol{\infty}}^{\mathbf{x}_t}$ serves as the lower bound of $\boldsymbol{\mathcal{B}}_{\mathrm{BE}}^{\mathbf{x}_t}$ when the PD position is perfectly known. In fact, a comparison between (66) and (67) reveals that $\boldsymbol{\mathcal{B}}_{\mathrm{BE}}^{\mathbf{x}_t}$ is more sensitive to $\mathbf{U}_t^i$ than $\boldsymbol{\mathcal{B}}_{\mathrm{BE}}^{\mathbf{r}_t^i}$, as $\mathbf{U}_t^i$ pertains solely to the PD. Specifically, when $\mathbf{U}_t^i$ approaches $\mathbf{0}$, (106) indicates that VLP fails, whereas (67) suggests that calibration is possible.

B. Asymptotic Limits of Closed-Form Balance State

(56) reveals that $\boldsymbol{\mathcal{B}}_{\mathrm{BE}}^{\mathbf{x}*}$ is influenced by both $\boldsymbol{\chi}$ and $\boldsymbol{\mathcal{L}}_{\mathrm{obs}}^{\mathbf{x}_t}$. The system parameters discussed in Subsection V-A all influence $\boldsymbol{\mathcal{L}}_{\mathrm{obs}}^{\mathbf{x}_t}$, so the following corollary will show their effects on $\boldsymbol{\mathcal{B}}_{\mathrm{BE}}^{\mathbf{x}*}$.

Corollary 9 (Asymptotic Analysis of Stable State): The four followings specify the asymptotic properties of the error $\boldsymbol{\mathcal{B}}_{\mathrm{BE}}^{\mathbf{x}*}$:

$$\lim_{\boldsymbol{\vartheta}\to 0}\boldsymbol{\mathcal{B}}_{\mathrm{BE}}^{\mathbf{x}*}=\boldsymbol{\infty},\ \lim_{\boldsymbol{\vartheta}\to\infty}\boldsymbol{\mathcal{B}}_{\mathrm{BE}}^{\mathbf{x}*}=\mathbf{0},\ \boldsymbol{\mathcal{B}}_{\mathrm{BE}}^{\mathbf{x}*}\propto\boldsymbol{\vartheta}^{-1},\quad (72)$$

$$\boldsymbol{\mathcal{B}}_{\mathrm{BE}}^{\mathbf{x}*}\propto(\rho_{t,\max}^{i,j})^4,\ \text{as}\ \rho_{t,\min}^{i,j}\to\infty,\quad (73)$$

$$\boldsymbol{\mathcal{B}}_{\mathrm{BE}}^{\mathbf{x}*}\propto\Omega_t^{-1},\ \text{since}\ \boldsymbol{\mathcal{L}}_{\mathrm{obs}}^{\mathbf{x}_t}\propto\Omega_t,\quad (74)$$

$$\lim_{\substack{\mathbf{U}_t^i\to\mathbf{0},\\ \forall i\in\Omega_t}}\boldsymbol{\mathcal{B}}_{\mathrm{BE}}^{\mathbf{x}*}=\boldsymbol{\infty},\ \lim_{\substack{\mathbf{U}_t^i\to\boldsymbol{\infty},\\ \forall i\in\Omega_t}}\boldsymbol{\mathcal{B}}_{\mathrm{BE}}^{\mathbf{x}*}=\boldsymbol{\mathcal{B}}_{\boldsymbol{\infty}}^{\mathbf{x}*},\ \boldsymbol{\mathcal{B}}_{\mathrm{BE}}^{\mathbf{x}*}\propto(\mathbf{U}_t^i)^{-1},\quad (75)$$

where $\boldsymbol{\mathcal{B}}_{\boldsymbol{\infty}}^{\mathbf{x}*}\in\mathbb{S}^3$ is the lower bound if $\mathbf{U}_t^i\to\boldsymbol{\infty}$, given by

$$\boldsymbol{\mathcal{B}}_{\boldsymbol{\infty}}^{\mathbf{x}*}=\frac{1}{2}\boldsymbol{\chi}^{-\frac{1}{2}}\left(\mathbf{I}_3+4\boldsymbol{\chi}^{\frac{1}{2}}\mathbf{V}(\mathbf{x}_t)^{-1}\boldsymbol{\chi}^{\frac{1}{2}}\right)^{\frac{1}{2}}\boldsymbol{\chi}^{-\frac{1}{2}}-\frac{1}{2}\boldsymbol{\chi}^{-1}.\quad (76)$$

Proof: (72)–(74) can be directly derived based on Corollary 5–7. Since $\mathcal{L}_{\text{obs}}^{\mathbf{x}_t} \to \mathbf{V}(\mathbf{x}_t)$ if $\mathbf{U}_t^i \to \infty$, we get (76) and then (75).

Remark 17: This elucidates the influence of these parameters, namely, $\boldsymbol{\vartheta}$, $\rho_t^{i,j}$, Ω_t and $\mathbf{U}_t^i$, on $\mathcal{B}_{\text{BE}}^{\mathbf{x}*}$. The corresponding impacts can be sequentially analogized to Corollary 5–8. The conclusions drawn are also similar to those in Remark 13–16.

VI. Numerical Results

In this section, we will conduct quantitative analyses from multiple perspectives to validate the theoretical correctness of the qualitative analysis of the convergence characteristics and asymptotic performance mentioned in Section IV and V.

A. Simulation Settings

While moving targets in real-world VLP scenarios are often irregular in shape, for the sake of simulation simplicity, we can model them as cuboids. We assume that a target, measuring 0.4 m × 0.4 m × 1.6 m, moves unpredictably and randomly within a room measuring 9 m × 9 m × 4 m. A total of 25 LED transmitters, each with a luminous efficacy of 100 lm/w, are mounted on the ceiling, spaced 2 m apart. The PD sensors are uniformly deployed on the floor, with nodes spaced at 0.25 m intervals. Without loss of generality, we set the order $d = 1$ and the DoF $\lambda = \gamma = 10$. Additionally, we have the following settings, i.e., $P_T = 5$ W, $A_R = 2$ cm^2, $F_R = 3.14$, $C_R = 8$, then the signal constant $\Psi_R = 40$ is assumed to isolate the effect of noise. For simplicity, ϖ is kept constant and equal to $\boldsymbol{\vartheta}$ in the simulation, allowing us to focus solely on $\boldsymbol{\Lambda}$ (i.e., $\boldsymbol{\Gamma}$), which affects $\boldsymbol{\vartheta}$. For instance, if $\boldsymbol{\Lambda} = 1/400$, then $\boldsymbol{\vartheta} = \varpi = 40$ dB. Unless otherwise specified, we set $\mathbf{U}_t^i = 10\mathbf{I}_3$ and $\boldsymbol{\chi} = 100\mathbf{I}_3$, without considering their effects on VLP performance.

B. Numerical Analysis

1) Error Evolution of Target Localization: Fig. 2 illustrates some physical quantities alongside their corresponding analytical solutions obtained during target localization under specific simulation conditions. It also highlights the performance gain, measured as the height difference between the peak and steady state values. This figure confirms that the VLP error, affected by PD position errors, fluctuates towards convergence, and that error $\mathcal{B}_{\text{BE}}^{\mathbf{x}_t}$ and $\mathcal{B}_{\text{BE}}^{\mathbf{x}*}$ share the same fluctuating upper and lower bounds, in accordance with Theorem 3–6 and Corollary 8–9. Additionally, Fig. 3 shows that the choice of mobile trajectory and reference LED sets the upper limit of $\mathcal{B}_{\text{BE}}^{\mathbf{x}_t}$ from the outset. By selecting an appropriate LED for positioning on the ideal trajectory, more optimal VLP performance can be achieved, with a relatively rapid convergence rate (see Remark 3).

2) The Impact of Noise Precision and Target Mobility: Fig. 4 reflects $\boldsymbol{\vartheta}$ through $\boldsymbol{\Lambda}$ and reveals that the error bound $\mathcal{B}_{\text{BE}}^{\mathbf{x}_t}$ and the stable state $\mathcal{B}_{\text{BE}}^{\mathbf{x}*}$ versus $\boldsymbol{\vartheta}$ and $\boldsymbol{\chi}$. The initial error (i.e., $\mathcal{B}_{\text{BE}}^{\mathbf{x}*,\max}$) remains consistent at the same $\boldsymbol{\vartheta}$ despite variations in $\boldsymbol{\chi}$, primarily due to $\mathcal{I}_{\text{pre}}^{\mathbf{x}_0} = \mathbf{0}$. A larger $\boldsymbol{\vartheta}$ and $\boldsymbol{\chi}$ will lead to better VLP performance, reduced stable state error, and slower final convergence rate (see Remark 11). Then, the arrangement variability of the blue and purple curves in Fig. 4, in relation to $\mathcal{B}_{\text{BE}}^{\mathbf{x}_t}$ and $\mathcal{B}_{\text{BE}}^{\mathbf{x}*}$, confirms the differing sensitivities of these

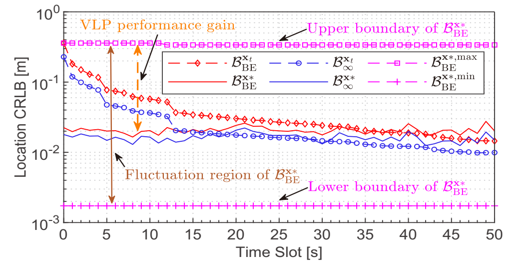

Fig. 2. A real-time CRLB example for a target of fixed size under the fixed mobile trajectory and reference LED ($\boldsymbol{\Lambda} = 1/400$, $\mathbf{U}_t^i = 10\mathbf{I}_3$, $\boldsymbol{\chi} = 100\mathbf{I}_3$).

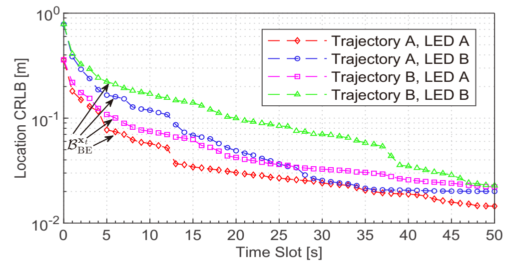

Fig. 3. Real-time CRLB for a measured target of fixed size over different mobile trajectories and reference LEDs ($\boldsymbol{\Lambda} = 1/400$, $\mathbf{U}_t^i = 10\mathbf{I}_3$, $\boldsymbol{\chi} = 100\mathbf{I}_3$).

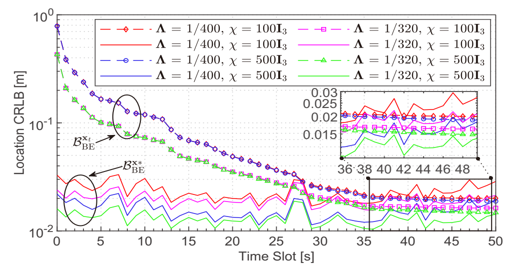

Fig. 4. Real-time CRLB for a fixed-size target over different noise accuracies and target mobilities under the fixed trajectory and LED ($\mathbf{U}_t^i = 10\mathbf{I}_3$).

variables to influencing factors. That is, $\mathcal{B}_{\text{BE}}^{\mathbf{x}_t}$ is more sensitive to $\boldsymbol{\vartheta}$, while $\mathcal{B}_{\text{BE}}^{\mathbf{x}*}$ is more responsive to $\boldsymbol{\chi}$, depending on the relevant parameters in (13) and (56). Similarly, just as $\mathcal{B}_{\text{BE}}^{\mathbf{x}_t}$ is lower bounded at infinite $\boldsymbol{\vartheta}$ as shown in Fig. 2, the PD position calibration error $\mathcal{B}_{\text{BE}}^{\mathbf{r}_t^i}$ follows a comparable pattern as depicted in Fig. 5, with its upper and lower bounds constrained by the limits at $\varpi = 0$ and $\varpi = \infty$, respectively (see Corollary 5). In addition, we validate the accuracy of the calibration CRLB by using root mean squared error (RMSE) based on maximum likelihood estimation (MLE) method. As shown in Fig. 5, the convergence trend obtained via MLE aligns with the pattern predicted by the CRLB, converging to higher error bounds at a faster rate. This highlights the intrinsic optimal performance of the CRLB, which will inevitably surpass any VLP algorithm.

3) The Impact of Reference Set Size and Transmission Distance: The transmission distance $\rho_t^{i,j}$ is entirely dependent on the actual locations of the positioning PD to be calibrated and the associated designated LED. When considering adjustments to the size of this measured target or the dimensions of this monitored room without changing the respective numbers of uniformly deployed LEDs and PDs, it appears that $\rho_t^{i,j}$ seem be affected by the modifications. However, in reality, what is

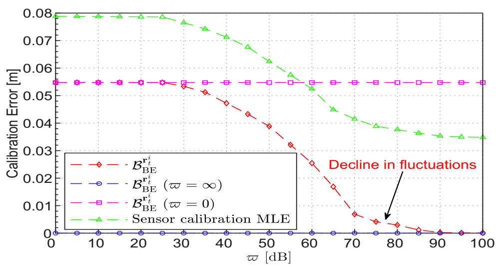

Fig. 5. Calibration error for the first positioning PD position used to locate the fixed-size target over different noise accuracies under the fixed mobile trajectory and reference LED at the initial moment ($\mathbf{U}_t^i = 10^3\mathbf{I}_3$, $\boldsymbol{\chi} = 100\mathbf{I}_3$).

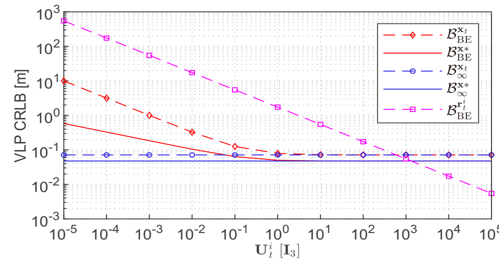

Fig. 8. Long-term location error ($t = 50$) and calibration error for the first positioning PD position used to locate the fixed-size target over different PD node precisions under the fixed trajectory and LED ($\mathbf{\Lambda} = 1/800$, $\boldsymbol{\chi} = 100\mathbf{I}_3$).

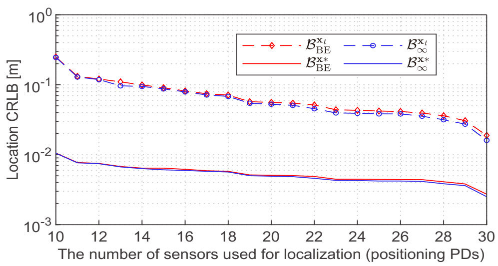

Fig. 6. Averaged initial CRLB (the overall mean value when $t = 0$) in the spatial domain for a fixed-size target over different PD sensor set sizes under the fixed reference LED ($\mathbf{\Lambda} = 1/320$, $\mathbf{U}_t^i = 10^3\mathbf{I}_3$, $\boldsymbol{\chi} = 10^3\mathbf{I}_3$).

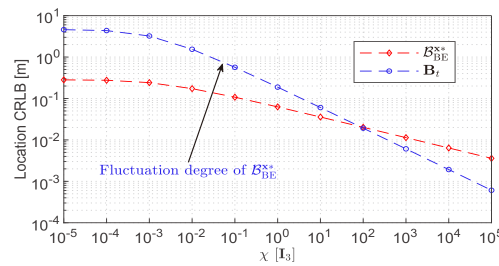

Fig. 9. Long-term location balance state $\mathcal{B}_{\text{BE}}^{\mathbf{x}*}$ ($t = 50$) and its fluctuation degree $\mathbf{B}_t$ for a target of fixed size over different transition precisions under the fixed mobile trajectory and reference LED ($\mathbf{\Lambda} = 1/400$, $\mathbf{U}_t^i = 10\mathbf{I}_3$).

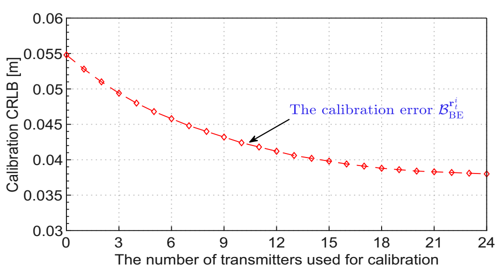

Fig. 7. Averaged initial spatial-domain calibration CRLB for the positioning PD position used to locate the fixed-size target over different transmitter set sizes under the fixed reference LED ($\mathbf{\Lambda} = 1/320$, $\mathbf{U}_t^i = 10^3\mathbf{I}_3$, $\boldsymbol{\chi} = 10^3\mathbf{I}_3$).

affected by these apparent alterations in transmission distance is the size of the respective reference sets during localization and calibration. For the sake of clarity, we will focus solely on discussing the effect of reference set size on VLP performance. To mitigate the impact of prediction information on VLP error, we then exclusively analyze the initial CRLB. By assessing the number of positioning PDs and associated errors across various target locations, we gauge the overall influence of the PD set size on VLP error through averaging, as shown in Fig. 6. The results suggest a direct correlation between VLP accuracy and reference set size, aligning with the findings depicted in Fig. 7. Here, the PD position calibration error decreases as the number of LEDs for calibration increases, as outlined in (65). It is worth noting that both Fig. 6 and 7 show that when the reference set sizes (i.e., Ω_t and Φ_t) reach a certain threshold, further addition of LOS observation information will no longer significantly improves VLP performance (see Remark 15).

4) The Impact of Location Uncertainties and Fluctuation Degree: The effect of PD position errors on VLP performance is illustrated in Fig. 8, indicating that the PD calibration error $\mathcal{B}_{\text{BE}}^{\mathbf{r}_t^i}$, target localization error $\mathcal{B}_{\text{BE}}^{\mathbf{x}_t}$, and its stable state $\mathcal{B}_{\text{BE}}^{\mathbf{x}*}$ all diminish with higher PD position accuracy $\mathbf{U}_t^i$. Since $\mathcal{B}_{\text{BE}}^{\mathbf{r}_t^i}$ is approximated as $(\mathbf{U}_t^i)^{-1}$ at low ϖ (cf. Fig. 5), the variation of $\mathcal{B}_{\text{BE}}^{\mathbf{r}_t^i}$ in Fig. 8 asymptotically follows a linear trend. In addition, $\mathcal{B}_{\text{BE}}^{\mathbf{x}_t}$ and $\mathcal{B}_{\text{BE}}^{\mathbf{x}*}$ asymptotically approach $\mathcal{B}_{\infty}^{\mathbf{x}_t}$ and $\mathcal{B}_{\infty}^{\mathbf{x}*}$ when $\mathbf{U}_t^i \to \infty$, respectively, which accords with Corollary 8 and (75). Recalling Theorem 5 and the definition of $\mathbf{B}_t$, the fluctuation degree of $\mathcal{B}_{\text{BE}}^{\mathbf{x}*}$ is heavily reliant on the transition precision $\boldsymbol{\chi}$, which notably influences $\mathcal{B}_{\text{BE}}^{\mathbf{x}*}$. Fig. 9 presents the relationship between $\mathcal{B}_{\text{BE}}^{\mathbf{x}*}$ and $\mathbf{B}_t$ for various values of $\boldsymbol{\chi}$, demonstrating that $\mathbf{B}_t$ responses well to the fluctuating pattern of $\mathcal{B}_{\text{BE}}^{\mathbf{x}*}$ and diminishes as $\boldsymbol{\chi}$ increases, consistent with Corollary 4.

5) The Impact of Sensor Deployment Density: As discussed in Remark 3 and 6, the optimal sensor deployment density is crucial for achieving a balance between the deployment cost and the localization performance. To examine the influence of different PD deployment densities on VLP performance, Fig. 10 compares VLP errors as a function of deployment density, using the same reference LED. The sensor deployment interval is varied from 0.25 m to 0.33 m, meaning denser and sparser deployment densities, respectively. While denser deployments clearly lead to better performance, the reference LED itself also impacts the performance gap (c.f. Fig. 3). Therefore, we further investigate the overall performance of passive VLP via multiple LEDs with sparser sensor deployment densities. The results manifest that, with the use of information from LEDs other than the reference LED, the performance with multiple LEDs is markedly superior to that of a single LED. Utilizing multiple LEDs for joint passive VLP is an effective strategy to compensate for the performance deficiencies associated with

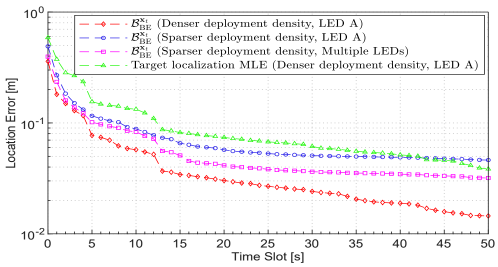

Fig. 10. Real-time location error for a measured target of fixed size over different sensor deployment densities under the fixed mobile trajectory ($\mathbf{\Lambda} =$ 1/400, $\mathbf{U}_t^i = 10\mathbf{I}_3$, $\boldsymbol{\chi} = 100\mathbf{I}_3$).

sparser sensor deployments. This approach mitigates the over-reliance on dense sensor networks, offering a crucial advantage of this multi-LED VLP system, in addition to its capability for calibration of the sensor position. Moreover, Fig. 10 presents the MLE-based RMSE results for the target location under the reference LED and the denser deployment density. It is evident that the gap between the MLE and CRLB gradually decreases over time, further validating the intrinsic accuracy of the error evolution principle outlined by the CRLB (see Theorem 1).

VII. Conclusion

This paper presents a comprehensive investigation into the performance limits of a passive VLP system based on multiple LED transmitters and PD sensors. This VLP scheme utilizes shadowing effects for device-free localization, depending on a single reference LED and multiple positioning PDs. Additionally, it employs numerous transmitters to calibrate the location error of each positioning PD sensor, ensuring optimal target localization performance. Specifically, the closed-form CRLB-based error bounds for mobile target localization and sensor position calibration are derived. The results give insights into how the observation information, impacted by various crucial physical parameters, can make informative contributions to the overall VLP performance through the intrinsic philosophies of localization cooperation. Additionally, the convergence properties of error evolution in both localization and calibration are investigated to reveal the long-term VLP performance decided by the intrinsic mechanism of this scheme. Finally, the validity and applicability of the derived error bounds are demonstrated by examining the asymptotic performance limits associated with key factors such as noise accuracy, transmission distance, reference set size and sensor position error. These closed-form CRLBs serve as a performance benchmark for actual passive VLP systems, providing not only a theoretical foundation for the design optimization of systematic localization algorithms but also practical insights for enhancing VLP performance.

In this paper, for analytical purposes, we suppose that all PDs are deployed on the floor, with all LEDs and PDs oriented strictly vertically, and the moving target is modeled as a regular cuboid. In future work, we aim to investigate how irregularly shaped targets can facilitate passive VLP using PDs distributed on walls, while focusing on correctly estimating the orientation of PDs amidst uncertainties in the orientation of multitudinous LEDs. Furthermore, we will develop a dedicated experimental setup for passive VLP to validate the robustness of localization algorithms optimized based on the derived CRLB. Our future efforts will address these challenges, significantly boosting the understanding of the intrinsic localization performance of this proposed passive VLP system across multiple dimensions.

Appendix A
Proof of Theorem 1

From the definition of $\boldsymbol{\alpha}_t$ in Section II-D, it is evident that $\boldsymbol{\alpha}_t$ is a large column vector composed of multiple spliced-together variables. We assume that each individual state variable within $\boldsymbol{\alpha}_t$ is independent of the others, and let $\boldsymbol{\alpha}_t^m$ denote the mth state variable in $\boldsymbol{\alpha}_t$. The BE-based FIM of $\boldsymbol{\alpha}_t$, denoted as $\boldsymbol{\mathcal{H}}_{\mathrm{BE}}^{\boldsymbol{\alpha}_t} \in \mathbb{S}$, is essential for obtaining the CRLB, given by $\boldsymbol{\mathcal{B}}_{\mathrm{BE}}^{\boldsymbol{\alpha}_t} = (\boldsymbol{\mathcal{H}}_{\mathrm{BE}}^{\boldsymbol{\alpha}_t})^{-1} \in \mathbb{S}$. Moreover, $\boldsymbol{\mathcal{H}}_{\mathrm{BE}}^{\boldsymbol{\alpha}_t}$ is determined by its constituent matrix elements $[\boldsymbol{\mathcal{H}}_{\mathrm{BE}}^{\boldsymbol{\alpha}_t}]_{m,n}$, where $[\bullet]_{m,n}$ denotes the (m,n)th element in the matrix. Based on (9), the critical and pivotal element $[\boldsymbol{\mathcal{H}}_{\mathrm{BE}}^{\boldsymbol{\alpha}_t}]_{m,n}$ can be structured as follows,

$$\begin{aligned}[\boldsymbol{\mathcal{H}}_{\mathrm{BE}}^{\boldsymbol{\alpha}_t}]_{m,n} &= -\mathbb{E}_{\mathbf{z}_t,\boldsymbol{\alpha}_t}\left\{\nabla_{\boldsymbol{\alpha}_t^m,\boldsymbol{\alpha}_t^{n\top}} \ln p(\boldsymbol{\alpha}_t|\mathbf{z}_{1:t})\right\} \\ &= \boldsymbol{\mathcal{I}}_{\mathrm{MLE}}^{\boldsymbol{\alpha}_t^m,\boldsymbol{\alpha}_t^n} \quad + \quad \iota_{m,n}\boldsymbol{\mathcal{I}}_{\mathrm{pre}}^{\boldsymbol{\alpha}_t^m} \\ &= \sum_{i\in\Omega_t^j}\left\{-\mathbb{E}_{\mathbf{z}_t,\boldsymbol{\alpha}_t}\{\nabla_{\boldsymbol{\alpha}_t^m,\boldsymbol{\alpha}_t^{n\top}} \ln p(\mathrm{z}_t^{i,j}|\boldsymbol{\alpha}_t)\}\right\} \\ &\quad + \iota_{m,n}\left\{-\mathbb{E}_{\mathbf{z}_t,\boldsymbol{\alpha}_t}\{\nabla_{\boldsymbol{\alpha}_t^m,\boldsymbol{\alpha}_t^{m\top}} \ln p(\boldsymbol{\alpha}_t^m|\mathbf{z}_{1:t-1})\}\right\}, \quad (77)\end{aligned}$$

where $\nabla_{\boldsymbol{\alpha}_t^m,\boldsymbol{\alpha}_t^{n\top}}(\bullet)$ denotes the second-order derivative obtained by deriving first $\boldsymbol{\alpha}_t^m$ and then $\boldsymbol{\alpha}_t^{n\top}$; $\boldsymbol{\mathcal{I}}_{\mathrm{MLE}}^{\boldsymbol{\alpha}_t^m,\boldsymbol{\alpha}_t^n}$ means the MLE-based FIM regarding $\forall\boldsymbol{\alpha}_t^m, \boldsymbol{\alpha}_t^n \in \{\boldsymbol{\alpha}_t\}$; $\boldsymbol{\mathcal{I}}_{\mathrm{pre}}^{\boldsymbol{\alpha}_t^m}$ defines the state prediction-based FIM associated with $\boldsymbol{\alpha}_t^m \in \{\boldsymbol{\alpha}_t\}$; and $\iota_{m,n}$ is used to ensure that $\boldsymbol{\mathcal{I}}_{\mathrm{pre}}^{\boldsymbol{\alpha}_t^m}$ exists solely as a diagonal matrix element, which is set as $\iota_{m,n} = 1$ only if $m = n$, while $\iota_{m,n} = 0$ otherwise. As the prediction information exists only with $\mathbf{x}_t$ and $\mathbf{r}_t^i$, then $\boldsymbol{\mathcal{I}}_{\mathrm{pre}}^{\mathbf{t}_t^j} = \mathbf{0}_{3\times 3}$ and $\boldsymbol{\mathcal{I}}_{\mathrm{pre}}^{\omega_t^{i,j}} = 0$ can be easily determined, while $\boldsymbol{\mathcal{I}}_{\mathrm{pre}}^{\mathbf{x}_t} \in \mathbb{S}^3$ and $\boldsymbol{\mathcal{I}}_{\mathrm{pre}}^{\mathbf{r}_t^i} \in \mathbb{S}^3$ regarding $\mathbf{x}_t$ and $\mathbf{r}_t^i$, respectively, can be given by

$$\boldsymbol{\mathcal{I}}_{\mathrm{pre}}^{\mathbf{x}_t} = -\mathbb{E}_{\mathbf{z}_t,\boldsymbol{\alpha}_t}\left\{\nabla_{\mathbf{x}_t,\mathbf{x}_t^\top} \ln \mathcal{N}(\mathbf{x}_t|\mathbf{x}_{t-1}^\flat, \boldsymbol{\Xi}_t^{\mathrm{pre}})\right\} = \boldsymbol{\Xi}_t^{\mathrm{pre}}, \quad (78)$$

$$\boldsymbol{\mathcal{I}}_{\mathrm{pre}}^{\mathbf{r}_t^i} = -\mathbb{E}_{\mathbf{z}_t,\boldsymbol{\alpha}_t}\left\{\nabla_{\mathbf{r}_t^i,\mathbf{r}_t^{i\top}} \ln \mathcal{N}(\mathbf{r}_t^i|\mathbf{s}_t^i, \mathbf{U}_t^i)\right\} = \mathbf{U}_t^i, \quad (79)$$

where $p(\boldsymbol{\alpha}_t^m|\mathbf{z}_{1:t-1})$ in the expression for $\boldsymbol{\mathcal{I}}_{\mathrm{pre}}^{\boldsymbol{\alpha}_t^m}$ in (77), and its counterpart in (78) and (79), can be given directly by (8).

For the Gaussian case, $\mathbb{E}\{\nabla_{\boldsymbol{\alpha}_t^m}(\bullet)\cdot\nabla_{\boldsymbol{\alpha}_t^{n\top}}(\bullet)\}$ is equivalent to $\mathbb{E}\{\nabla_{\boldsymbol{\alpha}_t^m,\boldsymbol{\alpha}_t^{n\top}}(\bullet)\}$ in solving the CRLB, where $\nabla_{\boldsymbol{\alpha}_t^m}(\bullet)$ is the first-order derivative. Thus, $\boldsymbol{\mathcal{I}}_{\mathrm{MLE}}^{\boldsymbol{\alpha}_t^m,\boldsymbol{\alpha}_t^n}$ can be given as

$$\begin{aligned}&\boldsymbol{\mathcal{I}}_{\mathrm{MLE}}^{\boldsymbol{\alpha}_t^m,\boldsymbol{\alpha}_t^n} \\ &= \sum_{i\in\Omega_t^j}\mathbb{E}_{\mathbf{z}_t,\boldsymbol{\alpha}_t}\left\{\nabla_{\boldsymbol{\alpha}_t^m}\ln p(\mathrm{z}_t^{i,j}|\boldsymbol{\alpha}_t)\nabla_{\boldsymbol{\alpha}_t^{n\top}}\ln p(\mathrm{z}_t^{i,j}|\boldsymbol{\alpha}_t)\right\} \\ &= \sum_{i\in\Omega_t^j}\mathbb{E}_{\boldsymbol{\alpha}_t}\left\{h_{\boldsymbol{\alpha}_t^m}^{i,j}(\omega_t^{i,j})^2\,\mathbb{E}_{\mathbf{z}_t}\left\{\left(\mathrm{z}_t^{i,j} - h(\mathbf{x}_t,\mathbf{r}_t^i,\mathbf{t}^j)\right)^2\right\}h_{\boldsymbol{\alpha}_t^n}^{i,j\,\top}\right\} \\ &= \sum_{i\in\Omega_t^j}\left\{\mathbb{E}_{\mathbf{x}_t,\mathbf{t}^j,\mathbf{r}_t^i}\{h_{\boldsymbol{\alpha}_t^m}^{i,j}\cdot h_{\boldsymbol{\alpha}_t^n}^{i,j\,\top}\}\cdot\mathbb{E}_{\omega_t^{i,j}}\{\omega_t^{i,j}\}\right\} \\ &= \sum_{i\in\Omega_t^j}\boldsymbol{\Lambda}\lambda\cdot h_{\boldsymbol{\alpha}_t^m}^{i,j}\cdot h_{\boldsymbol{\alpha}_t^n}^{i,j\,\top}, \quad (80)\end{aligned}$$

where $h^{i,j}_{\boldsymbol{\alpha}^m_t}$ denotes $\nabla_{\boldsymbol{\alpha}^m_t} h(\mathbf{x}_t, \mathbf{r}^i_t, \mathbf{t}^j)$, namely, the first-order derivative of $h(\mathbf{x}_t, \mathbf{r}^i_t, \mathbf{t}^j)$ with respect to α^m_t, and $\boldsymbol{\Lambda}\lambda \in \mathbb{R}$ is the noise precision expectation given by (6).

Derivation of $h^{i,j}_{\boldsymbol{\alpha}^m_t}$: Firstly, $h(\mathbf{x}_t, \mathbf{r}^i_t, \mathbf{t}^j)$ shown in (3) can be rewritten as $h(\mathbf{x}_t, \mathbf{r}^i_t, \mathbf{t}^j) = -\Psi_R(\rho^{i,j}_t)^{-2}\mathbf{g}^{i,j}_t(\mathbf{e}^{i,j}_{\mathbf{x},t})^\top \mathbf{u}^i$, where $\mathbf{g}^{i,j}_t(\mathbf{e}^{i,j}_{\mathbf{x},t}) \in \mathbb{R}^3$ is obtained by replacing all $\mathbf{e}^{i,j}_{\mathbf{r},t}$ in (31) with $\mathbf{e}^{i,j}_{\mathbf{x},t}$. Thus, $h^{i,j}_{\mathbf{x}_t} \in \mathbb{R}^3$ can be derived by (20) to (22) as

$$h^{i,j}_{\mathbf{x}_t} = -\Psi_R(\rho^{i,j}_t)^{-2}\mathbf{f}^{i,j}(\mathbf{x}_t)\mathbf{u}^i. \tag{81}$$

Secondly, based on (27) to (31), $h^{i,j}_{\mathbf{r}^i_t} \in \mathbb{R}^3$ is defined as

$$h^{i,j}_{\mathbf{r}^i_t} = -\Psi_R(\rho^{i,j}_t)^{-2}\mathbf{q}^{i,j}(\mathbf{r}^i_t)\mathbf{u}^i. \tag{82}$$

Thirdly, let $\mathbf{e}^{i,j}_{\mathbf{t},t} = -\mathbf{e}^{i,j}_{\mathbf{r},t} \in \mathbb{R}^3$, and then $h(\mathbf{x}_t, \mathbf{r}^i_t, \mathbf{t}^j)$ can be rewritten as $h(\mathbf{x}_t, \mathbf{r}^i_t, \mathbf{t}^j) = \Psi_R(\rho^{i,j}_t)^{-2}(-1)^d\mathbf{g}^{i,j}_t(\mathbf{e}^{i,j}_{\mathbf{t},t})^\top \mathbf{u}^i$. So, according to (35), $h^{i,j}_{\mathbf{t}^j} \in \mathbb{R}^3$ can be expressed as

$$h^{i,j}_{\mathbf{t}^j} = -\Psi_R(\rho^{i,j}_t)^{-2}\mathbf{j}^{i,j}(\mathbf{t}^j)\mathbf{u}^i. \tag{83}$$

Besides, $h^{i,j}_{\mathbf{r}^{i+}_t} = \mathbf{0}_{3\times 1}$ and $h^{i,j}_{\omega^{i,j}_t} = 0$ are obvious, in which $\mathbf{r}^{i+}_t$ means $\mathbf{r}^{i+}_t \neq \mathbf{r}^i_t$, i.e., the function contains $\mathbf{r}^i_t$ but not $\mathbf{r}^{i+}_t$.

Derivation of $\boldsymbol{\mathcal{I}}^{\alpha^m_t,\alpha^n_t}_{\text{MLE}}$: By (80), $\boldsymbol{\mathcal{I}}^{\mathbf{x}_t,\mathbf{x}_t}_{\text{MLE}} \in \mathbb{R}^{3\times 3}$ is given by

$$\begin{aligned}\boldsymbol{\mathcal{I}}^{\mathbf{x}_t,\mathbf{x}_t}_{\text{MLE}} &= \sum_{i\in\Omega^j_t} \underbrace{\boldsymbol{\Lambda}\lambda\Psi^2_R(\rho^{i,j}_t)^{-4}}_{\boldsymbol{\vartheta}}\mathbf{f}^{i,j}(\mathbf{x}_t)\mathbf{u}^i\mathbf{u}^{i\top}\mathbf{f}^{i,j}(\mathbf{x}_t)^\top \\ &= \boldsymbol{\vartheta}\cdot\mathbf{F}(\mathbf{x}_t)\mathbf{K}(\mathbf{u})\mathbf{K}(\mathbf{u})^\top\mathbf{F}(\mathbf{x}_t)^\top,\end{aligned} \tag{84}$$

where $\mathbf{F}(\mathbf{x}_t)$ and $\mathbf{K}(\mathbf{u})$ are denoted as (19) and (23), respectively. Similarly, $\boldsymbol{\mathcal{I}}^{\mathbf{x}_t,\mathbf{t}^j}_{\text{MLE}} \in \mathbb{R}^{3\times 3}$ is easily got from (80) as

$$\boldsymbol{\mathcal{I}}^{\mathbf{x}_t,\mathbf{t}^j}_{\text{MLE}} = \boldsymbol{\vartheta}\cdot\mathbf{F}(\mathbf{x}_t)\mathbf{K}(\mathbf{u})\mathbf{K}(\mathbf{u})^\top\mathbf{J}(\mathbf{t}^j)^\top, \tag{85}$$

where $\mathbf{J}(\mathbf{t}^j)$ is defined as (34). When either $\boldsymbol{\alpha}^m_t$ and $\boldsymbol{\alpha}^n_t$ is $\mathbf{r}^i_t$, $\sum$ in (80) will be invalidated, as there is only a unique function can provide the relevant $\mathbf{r}^i_t$. So, $\boldsymbol{\mathcal{I}}^{\mathbf{x}_t,\mathbf{r}^i_t}_{\text{MLE}}$, $\boldsymbol{\mathcal{I}}^{\mathbf{t}^j,\mathbf{r}^i_t}_{\text{MLE}}$ and $\boldsymbol{\mathcal{I}}^{\mathbf{r}^i_t,\mathbf{r}^i_t}_{\text{MLE}}$, which all belong to $\mathbb{R}^{3\times 3}$, can be expressed as follows, respectively,

$$\boldsymbol{\mathcal{I}}^{\mathbf{x}_t,\mathbf{r}^i_t}_{\text{MLE}} = \boldsymbol{\Lambda}\lambda\cdot h^{i,j}_{\mathbf{x}_t}\cdot h^{i,j\top}_{\mathbf{r}^i_t} = \boldsymbol{\vartheta}\cdot\mathbf{f}(\mathbf{x}_t)\mathbf{u}^i\mathbf{u}^{i\top}\mathbf{q}(\mathbf{r}^i_t)^\top, \tag{86}$$

$$\boldsymbol{\mathcal{I}}^{\mathbf{t}^j,\mathbf{r}^i_t}_{\text{MLE}} = \boldsymbol{\Lambda}\lambda\cdot h^{i,j}_{\mathbf{t}^j}\cdot h^{i,j\top}_{\mathbf{r}^i_t} = \boldsymbol{\vartheta}\cdot\mathbf{j}(\mathbf{t}^j)\mathbf{u}^i\mathbf{u}^{i\top}\mathbf{q}(\mathbf{r}^i_t)^\top, \tag{87}$$

$$\boldsymbol{\mathcal{I}}^{\mathbf{r}^i_t,\mathbf{r}^i_t}_{\text{MLE}} = \boldsymbol{\Lambda}\lambda\cdot h^{i,j}_{\mathbf{r}^i_t}\cdot h^{i,j\top}_{\mathbf{r}^i_t} = \boldsymbol{\vartheta}\cdot\mathbf{q}(\mathbf{r}^i_t)\mathbf{u}^i\mathbf{u}^{i\top}\mathbf{q}(\mathbf{r}^i_t)^\top, \tag{88}$$

where $\mathbf{f}(\mathbf{x}_t)$, $\mathbf{q}(\mathbf{r}^i_t)$ and $\mathbf{j}(\mathbf{t}^j)$ can be given by (25), (26) and (37), respectively. As for $\boldsymbol{\mathcal{I}}^{\mathbf{t}^j,\mathbf{t}^j}_{\text{MLE}} \in \mathbb{R}^{3\times 3}$, it can be formed as

$$\boldsymbol{\mathcal{I}}^{\mathbf{t}^j,\mathbf{t}^j}_{\text{MLE}} = \boldsymbol{\vartheta}\cdot\mathbf{J}(\mathbf{t}^j)\mathbf{K}(\mathbf{u})\mathbf{K}(\mathbf{u})^\top\mathbf{J}(\mathbf{t}^j)^\top. \tag{89}$$

Since $h^{i,j}_{\omega^{i,j}_t} = 0$, $\boldsymbol{\mathcal{I}}^{\mathbf{x}_t,\omega^{i,j}_t}_{\text{MLE}} = \boldsymbol{\mathcal{I}}^{\mathbf{t}^j,\omega^{i,j}_t}_{\text{MLE}} = \boldsymbol{\mathcal{I}}^{\mathbf{r}^i_t,\omega^{i,j}_t}_{\text{MLE}} = \mathbf{0}_{3\times 1}$ and $\boldsymbol{\mathcal{I}}^{\omega^{i,j}_t,\omega^{i,j}_t}_{\text{MLE}} = \boldsymbol{\mathcal{I}}^{\omega^{i,j}_t,\omega^{i+,j}_t}_{\text{MLE}} = 0$, where $\omega^{i+,j}_t$ means $\omega^{i+,j}_t \neq \omega^{i,j}_t$. Additionally, as $h^{i,j}_{\mathbf{r}^{i+}_t} = \mathbf{0}_{3\times 1}$, we can get $\boldsymbol{\mathcal{I}}^{\mathbf{r}^i_t,\mathbf{r}^{i+}_t}_{\text{MLE}} = \mathbf{0}_{3\times 3}$.

Derivation of $\boldsymbol{\mathcal{H}}^{\boldsymbol{\alpha}_t}_{\text{BE}}$: Using $\boldsymbol{\mathcal{I}}^{\alpha^m_t}_{\text{pre}}$ and $\boldsymbol{\mathcal{I}}^{\alpha^m_t,\alpha^n_t}_{\text{MLE}}$ gained above, five block matrices can be defined as shown below, namely,

$$\mathbf{A}^{\mathbf{x}_t}_{\text{BE}} = \begin{bmatrix}\boldsymbol{\mathcal{I}}^{\mathbf{x}_t,\mathbf{x}_t}_{\text{MLE}} + \boldsymbol{\mathcal{I}}^{\mathbf{x}_t}_{\text{pre}} & \boldsymbol{\mathcal{I}}^{\mathbf{x}_t,\mathbf{t}^j}_{\text{MLE}} \\ \boldsymbol{\mathcal{I}}^{\mathbf{t}^j,\mathbf{x}_t}_{\text{MLE}} & \boldsymbol{\mathcal{I}}^{\mathbf{t}^j,\mathbf{t}^j}_{\text{MLE}} + \boldsymbol{\mathcal{I}}^{\mathbf{t}^j}_{\text{pre}}\end{bmatrix} = \begin{bmatrix}\boldsymbol{\mathcal{I}}^{\mathbf{x}_t,\mathbf{x}_t}_{\text{BE}} & \boldsymbol{\mathcal{I}}^{\mathbf{x}_t,\mathbf{t}^j}_{\text{MLE}} \\ \boldsymbol{\mathcal{I}}^{\mathbf{t}^j,\mathbf{x}_t}_{\text{MLE}} & \boldsymbol{\mathcal{I}}^{\mathbf{t}^j,\mathbf{t}^j}_{\text{MLE}}\end{bmatrix}, \tag{90}$$

$$\mathbf{B}^{\mathbf{x}_t}_{\text{BE}} = \begin{bmatrix}\boldsymbol{\mathcal{I}}^{\mathbf{x}_t,\mathbf{r}^1_t}_{\text{MLE}} & \cdots & \boldsymbol{\mathcal{I}}^{\mathbf{x}_t,\mathbf{r}^{\Omega_t}_t}_{\text{MLE}} \\ \boldsymbol{\mathcal{I}}^{\mathbf{t}^j,\mathbf{r}^1_t}_{\text{MLE}} & \cdots & \boldsymbol{\mathcal{I}}^{\mathbf{t}^j,\mathbf{r}^{\Omega_t}_t}_{\text{MLE}}\end{bmatrix}, \tag{91}$$

$$\begin{aligned}\mathbf{C}^{\mathbf{x}_t}_{\text{BE}} &= \begin{bmatrix}\boldsymbol{\mathcal{I}}^{\mathbf{r}^1_t,\mathbf{r}^1_t}_{\text{MLE}} + \boldsymbol{\mathcal{I}}^{\mathbf{r}^1_t}_{\text{pre}} & \cdots & \boldsymbol{\mathcal{I}}^{\mathbf{r}^1_t,\mathbf{r}^{\Omega_t}_t}_{\text{MLE}} \\ \vdots & \ddots & \vdots \\ \boldsymbol{\mathcal{I}}^{\mathbf{r}^{\Omega_t}_t,\mathbf{r}^1_t}_{\text{MLE}} & \cdots & \boldsymbol{\mathcal{I}}^{\mathbf{r}^{\Omega_t}_t,\mathbf{r}^{\Omega_t}_t}_{\text{MLE}} + \boldsymbol{\mathcal{I}}^{\mathbf{r}^{\Omega_t}_t}_{\text{pre}}\end{bmatrix} \\ &= diag(\boldsymbol{\mathcal{I}}^{\mathbf{r}^1_t,\mathbf{r}^1_t}_{\text{MLE}} + \boldsymbol{\mathcal{I}}^{\mathbf{r}^1_t}_{\text{pre}}, \cdots, \boldsymbol{\mathcal{I}}^{\mathbf{r}^{\Omega_t}_t,\mathbf{r}^{\Omega_t}_t}_{\text{MLE}} + \boldsymbol{\mathcal{I}}^{\mathbf{r}^{\Omega_t}_t}_{\text{pre}}),\end{aligned} \tag{92}$$

$$\mathbf{D}^{\mathbf{x}_t}_{\text{BE}} = \begin{bmatrix}\boldsymbol{\mathcal{I}}^{\mathbf{x}_t,\omega^{1,j}_t}_{\text{MLE}} & \cdots & \boldsymbol{\mathcal{I}}^{\mathbf{x}_t,\omega^{\Omega_t,j}_t}_{\text{MLE}} \\ \vdots & \ddots & \vdots \\ \boldsymbol{\mathcal{I}}^{\mathbf{r}^{\Omega_t}_t,\omega^{1,j}_t}_{\text{MLE}} & \cdots & \boldsymbol{\mathcal{I}}^{\mathbf{r}^{\Omega_t}_t,\omega^{\Omega_t,j}_t}_{\text{MLE}}\end{bmatrix} = \mathbf{0}_{3(\Omega_t+2)\times\Omega_t}, \tag{93}$$

$$\begin{aligned}\mathbf{E}^{\mathbf{x}_t}_{\text{BE}} &= \begin{bmatrix}\boldsymbol{\mathcal{I}}^{\omega^{1,j}_t,\omega^{1,j}_t}_{\text{MLE}} + \boldsymbol{\mathcal{I}}^{\omega^{1,j}_t}_{\text{pre}} & \cdots & \boldsymbol{\mathcal{I}}^{\omega^{1,j}_t,\omega^{\Omega_t,j}_t}_{\text{MLE}} \\ \vdots & \ddots & \vdots \\ \boldsymbol{\mathcal{I}}^{\omega^{\Omega_t,j}_t,\omega^{1,j}_t}_{\text{MLE}} & \cdots & \boldsymbol{\mathcal{I}}^{\omega^{\Omega_t,j}_t,\omega^{\Omega_t,j}_t}_{\text{MLE}} + \boldsymbol{\mathcal{I}}^{\omega^{\Omega_t,j}_t}_{\text{pre}}\end{bmatrix} \\ &= \mathbf{0}_{\Omega_t\times\Omega_t},\end{aligned} \tag{94}$$

where $\boldsymbol{\mathcal{I}}^{\mathbf{x}_t,\mathbf{x}_t}_{\text{BE}} \in \mathbb{R}^{3\times 3}$ is a simplification of $\boldsymbol{\mathcal{I}}^{\mathbf{x}_t,\mathbf{x}_t}_{\text{MLE}} + \boldsymbol{\mathcal{I}}^{\mathbf{x}_t}_{\text{pre}}$, and $\boldsymbol{\mathcal{I}}^{\boldsymbol{\alpha}^m_t,\boldsymbol{\alpha}^n_t}_{\text{MLE}} = (\boldsymbol{\mathcal{I}}^{\boldsymbol{\alpha}^n_t,\boldsymbol{\alpha}^m_t}_{\text{MLE}})^\top$ holds when continuously differentiable.

Hence, the final structure of $\boldsymbol{\mathcal{H}}^{\boldsymbol{\alpha}_t}_{\text{BE}}$ can be constructed as

$$\boldsymbol{\mathcal{H}}^{\boldsymbol{\alpha}_t}_{\text{BE}} = \begin{bmatrix}\begin{bmatrix}\mathbf{A}^{\mathbf{x}_t}_{\text{BE}} & \mathbf{B}^{\mathbf{x}_t}_{\text{BE}} \\ (\mathbf{B}^{\mathbf{x}_t}_{\text{BE}})^\top & \mathbf{C}^{\mathbf{x}_t}_{\text{BE}}\end{bmatrix} & \mathbf{D}^{\mathbf{x}_t}_{\text{BE}} \\ (\mathbf{D}^{\mathbf{x}_t}_{\text{BE}})^\top & \mathbf{E}^{\mathbf{x}_t}_{\text{BE}}\end{bmatrix}, \tag{95}$$

where $\boldsymbol{\mathcal{B}}^{\mathbf{x}_t}_{\text{BE}} = [(\boldsymbol{\mathcal{H}}^{\boldsymbol{\alpha}_t}_{\text{BE}})^{-1}]_{[1:3,1:3]}$ can be then obtained, while $[\bullet]_{[1:a,1:a]}$ denotes the upper left $a\times a$ submatrix of the matrix.

Then, applying the Schur complement [44], $\boldsymbol{\mathcal{B}}^{\mathbf{x}_t}_{\text{BE}}$, which has been given great attention, can be specifically described as

$$\begin{aligned}\boldsymbol{\mathcal{B}}^{\mathbf{x}_t}_{\text{BE}} &= \left[\begin{bmatrix}\mathbf{A}^{\mathbf{x}_t}_{\text{BE}} & \mathbf{B}^{\mathbf{x}_t}_{\text{BE}} \\ (\mathbf{B}^{\mathbf{x}_t}_{\text{BE}})^\top & \mathbf{C}^{\mathbf{x}_t}_{\text{BE}}\end{bmatrix}^{-1}\right]_{[1:3,1:3]} \\ &= \left[\left(\mathbf{A}^{\mathbf{x}_t}_{\text{BE}} - \mathbf{B}^{\mathbf{x}_t}_{\text{BE}}(\mathbf{C}^{\mathbf{x}_t}_{\text{BE}})^{-1}(\mathbf{B}^{\mathbf{x}_t}_{\text{BE}})^\top\right)^{-1}\right]_{[1:3,1:3]} \\ &= \left[\begin{bmatrix}\mathbf{A}^{\mathbf{x}_t}_{\sharp\text{BE}} & \mathbf{B}^{\mathbf{x}_t}_{\sharp\text{BE}} \\ \mathbf{D}^{\mathbf{x}_t}_{\sharp\text{BE}} & \mathbf{C}^{\mathbf{x}_t}_{\sharp\text{BE}}\end{bmatrix}^{-1}\right]_{[1:3,1:3]} \\ &= \left(\mathbf{A}^{\mathbf{x}_t}_{\sharp\text{BE}} - \mathbf{B}^{\mathbf{x}_t}_{\sharp\text{BE}}(\mathbf{C}^{\mathbf{x}_t}_{\sharp\text{BE}})^{-1}\mathbf{D}^{\mathbf{x}_t}_{\sharp\text{BE}}\right)^{-1},\end{aligned} \tag{96}$$

where $\mathbf{A}^{\mathbf{x}_t}_{\sharp\text{BE}}$, $\mathbf{B}^{\mathbf{x}_t}_{\sharp\text{BE}}$, $\mathbf{C}^{\mathbf{x}_t}_{\sharp\text{BE}}$ and $\mathbf{D}^{\mathbf{x}_t}_{\sharp\text{BE}}$ are cast, respectively, as

$$\mathbf{A}^{\mathbf{x}_t}_{\sharp\text{BE}} = \boldsymbol{\mathcal{I}}^{\mathbf{x}_t,\mathbf{x}_t}_{\text{BE}} - \sum_{i\in\Omega^j_t}\boldsymbol{\mathcal{I}}^{\mathbf{x}_t,\mathbf{r}^i_t}_{\text{MLE}}(\boldsymbol{\mathcal{I}}^{\mathbf{r}^i_t,\mathbf{r}^i_t}_{\text{MLE}} + \boldsymbol{\mathcal{I}}^{\mathbf{r}^i_t}_{\text{pre}})^{-1}(\boldsymbol{\mathcal{I}}^{\mathbf{x}_t,\mathbf{r}^i_t}_{\text{MLE}})^\top, \tag{97}$$

$$\mathbf{B}^{\mathbf{x}_t}_{\sharp\text{BE}} = \boldsymbol{\mathcal{I}}^{\mathbf{x}_t,\mathbf{t}^j}_{\text{MLE}} - \sum_{i\in\Omega^j_t}\boldsymbol{\mathcal{I}}^{\mathbf{x}_t,\mathbf{r}^i_t}_{\text{MLE}}(\boldsymbol{\mathcal{I}}^{\mathbf{r}^i_t,\mathbf{r}^i_t}_{\text{MLE}} + \boldsymbol{\mathcal{I}}^{\mathbf{r}^i_t}_{\text{pre}})^{-1}(\boldsymbol{\mathcal{I}}^{\mathbf{t}^j,\mathbf{r}^i_t}_{\text{MLE}})^\top, \tag{98}$$

$$\mathbf{C}^{\mathbf{x}_t}_{\sharp\text{BE}} = \boldsymbol{\mathcal{I}}^{\mathbf{t}^j,\mathbf{t}^j}_{\text{MLE}} - \sum_{i\in\Omega^j_t}\boldsymbol{\mathcal{I}}^{\mathbf{t}^j,\mathbf{r}^i_t}_{\text{MLE}}(\boldsymbol{\mathcal{I}}^{\mathbf{r}^i_t,\mathbf{r}^i_t}_{\text{MLE}} + \boldsymbol{\mathcal{I}}^{\mathbf{r}^i_t}_{\text{pre}})^{-1}(\boldsymbol{\mathcal{I}}^{\mathbf{t}^j,\mathbf{r}^i_t}_{\text{MLE}})^\top, \tag{99}$$

$$\mathbf{D}^{\mathbf{x}_t}_{\sharp\text{BE}} = \boldsymbol{\mathcal{I}}^{\mathbf{t}^j,\mathbf{x}_t}_{\text{MLE}} - \sum_{i\in\Omega^j_t}\boldsymbol{\mathcal{I}}^{\mathbf{t}^j,\mathbf{r}^i_t}_{\text{MLE}}(\boldsymbol{\mathcal{I}}^{\mathbf{r}^i_t,\mathbf{r}^i_t}_{\text{MLE}} + \boldsymbol{\mathcal{I}}^{\mathbf{r}^i_t}_{\text{pre}})^{-1}(\boldsymbol{\mathcal{I}}^{\mathbf{x}_t,\mathbf{r}^i_t}_{\text{MLE}})^\top, \tag{100}$$

where these functions in $\mathbb{R}^{3\times 3}$ can be used to define $\mathcal{H}_{\text{BE}}^{\mathbf{x}_t}$ as

$$\begin{aligned}\mathcal{H}_{\text{BE}}^{\mathbf{x}_t} &= \mathbf{A}_{\sharp\text{BE}}^{\mathbf{x}_t} - \mathbf{B}_{\sharp\text{BE}}^{\mathbf{x}_t}(\mathbf{C}_{\sharp\text{BE}}^{\mathbf{x}_t})^{-1}\mathbf{D}_{\sharp\text{BE}}^{\mathbf{x}_t}\\ &= \mathcal{I}_{\text{MLE}}^{\mathbf{x}_t,\mathbf{x}_t} - \sum_{i\in\Omega_t^j}\mathcal{I}_{\text{MLE}}^{\mathbf{x}_t,\mathbf{r}_t^i}(\mathcal{I}_{\text{MLE}}^{\mathbf{r}_t^i,\mathbf{r}_t^i}+\mathcal{I}_{\text{pre}}^{\mathbf{r}_t^i})^{-1}(\mathcal{I}_{\text{MLE}}^{\mathbf{x}_t,\mathbf{r}_t^i})^\top\\ &\quad - \mathbf{B}_{\sharp\text{BE}}^{\mathbf{x}_t}(\mathbf{C}_{\sharp\text{BE}}^{\mathbf{x}_t})^{-1}\mathbf{D}_{\sharp\text{BE}}^{\mathbf{x}_t} + \mathcal{I}_{\text{pre}}^{\mathbf{x}_t}\\ &= \mathcal{L}_{\text{obs}}^{\mathbf{x}_t} + (\chi^{-1}+\mathcal{B}_{\text{BE}}^{\mathbf{x}_{t-1}})^{-1},\end{aligned} \tag{101}$$

where $\mathcal{L}_{\text{obs}}^{\mathbf{x}_t}$ remains a complex matrix requiring matrixization to refine its expression. Then, based on (14), the complicated $\mathcal{L}_{\text{obs}}^{\mathbf{x}_t}$ is denoted as $\mathcal{L}_{\text{obs}}^{\mathbf{x}_t} = \vartheta\cdot\mathcal{O}_{\text{BE}}^{\mathbf{x}_t}$. Theorem 1 is proved.

Appendix B
Proof of Theorem 3

Based on Corollary 1, $\mathcal{L}_{\text{obs}}^{\mathbf{x}_t} \succeq \mathcal{J}_{\mathbf{x}_t}$ is equivalent to $\mathcal{L}_{\text{obs}}^{\mathbf{x}_t} + \mathcal{I}_{\text{pre}}^{\mathbf{x}_t} \succeq \mathcal{H}_{\text{BE}}^{\mathbf{x}_{t-1}}$, thus we get $\mathcal{H}_{\text{BE}}^{\mathbf{x}_t} \succeq \mathcal{H}_{\text{BE}}^{\mathbf{x}_{t-1}}$ from (13), and then $\mathcal{B}_{\text{BE}}^{\mathbf{x}_t} \preceq \mathcal{B}_{\text{BE}}^{\mathbf{x}_{t-1}}$ holds. As $\mathcal{B}_{\text{BE}}^{\mathbf{x}_t} = (\mathcal{L}_{\text{obs}}^{\mathbf{x}_t}+\mathcal{I}_{\text{pre}}^{\mathbf{x}_t})^{-1}$, if $\mathcal{B}_{\text{BE}}^{\mathbf{x}_t} \preceq \mathcal{B}_{\text{BE}}^{\mathbf{x}_{t-1}}$ holds and also non-decreasing $\mathcal{L}_{\text{obs}}^{\mathbf{x}_t}$ is satisfied, we can undoubtedly have $\mathcal{B}_{\text{BE}}^{\mathbf{x}_{t+1}} \preceq \mathcal{B}_{\text{BE}}^{\mathbf{x}_t}$. Theorem 3 is proved.

Appendix C
Proof of Theorem 4

Based on the matrix inversion lemma, we can have $\mathcal{B}_{\text{BE}}^{\mathbf{x}_t} - \mathcal{B}_{\text{BE}}^{\mathbf{x}_{t+1}} = \mathcal{B}_{\text{BE}}^{\mathbf{x}_{t-1}} - \mathcal{B}_{\text{BE}}^{\mathbf{x}_t} - \mathcal{K}_{\mathbf{x}_t}$, where $\mathcal{K}_{\mathbf{x}_t}$ is given by

$$\begin{aligned}\mathcal{K}_{\mathbf{x}_t}\in\mathbb{S}^3 = &\Big((\chi^{-1}+\mathcal{B}_{\text{BE}}^{\mathbf{x}_{t-1}})^{-1}(\mathcal{L}_{\text{obs}}^{\mathbf{x}_t})^{-1}(\chi^{-1}+\mathcal{B}_{\text{BE}}^{\mathbf{x}_{t-1}})^{-1}\\ &+(\chi^{-1}+\mathcal{B}_{\text{BE}}^{\mathbf{x}_{t-1}})^{-1}\Big)^{-1}\\ &-\Big((\chi^{-1}+\mathcal{B}_{\text{BE}}^{\mathbf{x}_t})^{-1}(\mathcal{L}_{\text{obs}}^{\mathbf{x}_{t+1}})^{-1}(\chi^{-1}+\mathcal{B}_{\text{BE}}^{\mathbf{x}_t})^{-1}\\ &+(\chi^{-1}+\mathcal{B}_{\text{BE}}^{\mathbf{x}_t})^{-1}\Big)^{-1},\end{aligned} \tag{102}$$

where $\mathcal{K}_{\mathbf{x}_t}\succeq\mathbf{0}$ and $\mathcal{K}_{\mathbf{x}_t}\to\mathbf{0}$ clearly hold. Then, $\zeta_{\mathbf{x}}$ in (53) is formed as $\zeta_{\mathbf{x}} = \lim_{t\to\infty}\frac{||\mathcal{K}_{\mathbf{x}_t}||_F}{||\mathcal{B}_{\text{BE}}^{\mathbf{x}_{t-1}}-\mathcal{B}_{\text{BE}}^{\mathbf{x}_t}||_F}$, since $\mathcal{K}_{\mathbf{x}_t}\succeq\mathbf{0}$. Also, (54) means the final simplification of $\zeta_{\mathbf{x}}$, achieved after extensive calculations, which will not be reiterated here. Thus, $\zeta_{\mathbf{x}},\eta_{\mathbf{x}}\in(0,1)$ can be then attained. Theorem 4 is proved.

Appendix D
Proof of Theorem 5

From the error convergence mechanism specified in Remark 8, it is clear that the dynamic balance between the information loss $\mathcal{J}_{\mathbf{x}_t}$ and the observation information $\mathcal{L}_{\text{obs}}^{\mathbf{x}_t}$ leads to a stable error $\mathcal{B}_{\text{BE}}^{\mathbf{x}*}$, where the central relational equation is represented by $\mathcal{L}_{\text{obs}}^{\mathbf{x}_t} = \mathcal{J}_{\mathbf{x}_t}$. This indicates the necessity for resolving the typical algebraic Riccati equation (ARE) [45] given as follows, i.e., $\mathcal{B}_{\text{BE}}^{\mathbf{x}*}\chi\mathcal{B}_{\text{BE}}^{\mathbf{x}*}+\mathcal{B}_{\text{BE}}^{\mathbf{x}*} = (\mathcal{L}_{\text{obs}}^{\mathbf{x}_t})^{-1}$ got from (51). In fact, (13) serves as the error evolution equation under matrix iteration, thereby allowing $\mathcal{B}_{\text{BE}}^{\mathbf{x}*}$ to be considered a fixed-point solution.

In order to solve the above ARE, first splitting χ, we have

$$\Big(\mathcal{B}_{\text{BE}}^{\mathbf{x}*}\chi^{\frac{1}{2}}+\frac{1}{2}\chi^{-\frac{1}{2}}\Big)\Big(\chi^{\frac{1}{2}}\mathcal{B}_{\text{BE}}^{\mathbf{x}*}+\frac{1}{2}\chi^{-\frac{1}{2}}\Big) = (\mathcal{L}_{\text{obs}}^{\mathbf{x}_t})^{-1}+\frac{1}{4}\chi^{-1}. \tag{103}$$

Premultiply and postmultiply $\chi^{\frac{1}{2}}$ at both sides of (103) as

$$\Big(\chi^{\frac{1}{2}}\mathcal{B}_{\text{BE}}^{\mathbf{x}*}\chi^{\frac{1}{2}}+\frac{1}{2}\mathbf{I}_3\Big)^2 = \chi^{\frac{1}{2}}(\mathcal{L}_{\text{obs}}^{\mathbf{x}_t})^{-1}\chi^{\frac{1}{2}}+\frac{1}{4}\mathbf{I}_3. \tag{104}$$

Hence, (56) can be obtained by squaring both sides of (104). Then, the relevant ARE can be solved. Theorem 5 is proved.

Appendix E
Proof of Theorem 6

Based on (13), the VLP error $\mathcal{B}_{\text{BE}}^{\mathbf{x}_t}$ asymptotically follows

$$\lim_{\mathcal{B}_{\text{BE}}^{\mathbf{x}_{t-1}}\to\mathbf{0}}\mathcal{B}_{\text{BE}}^{\mathbf{x}_t} = (\mathcal{L}_{\text{obs}}^{\mathbf{x}_t}+\chi)^{-1},\ \lim_{\mathcal{B}_{\text{BE}}^{\mathbf{x}_{t-1}}\to\infty}\mathcal{B}_{\text{BE}}^{\mathbf{x}_t} = (\mathcal{L}_{\text{obs}}^{\mathbf{x}_t})^{-1}, \tag{105}$$

where $(\mathcal{L}_{\text{obs}}^{\mathbf{x}_t}+\chi)^{-1}\preceq\mathcal{B}_{\text{BE}}^{\mathbf{x}_t}\preceq(\mathcal{L}_{\text{obs}}^{\mathbf{x}_t})^{-1}$ holds. In fact, the observation information $\mathcal{L}_{\text{obs}}^{\mathbf{x}_t}$ may vary within a bounded area, e.g., $\mathcal{L}_{\text{obs}}^{\mathbf{x}_t}\in[\mathcal{L}_{\text{obs}}^{\mathbf{x},\min},\mathcal{L}_{\text{obs}}^{\mathbf{x},\max}]$. Then, $\mathcal{B}_{\text{BE}}^{\mathbf{x}_t}$ gradually decreases and ultimately fluctuates within a region. Without delving into $\mathcal{B}_{\text{BE}}^{\mathbf{x}*}$ denoted in (56), then the bound of $\mathcal{B}_{\text{BE}}^{\mathbf{x}*}$ can be assumed to depend on the bound of $\mathcal{B}_{\text{BE}}^{\mathbf{x}_t}$. Theorem 6 is proved.

Appendix F
Proof of Corollary 6

We know $\rho_{t,\max}^{i,j}$ and $\rho_{t,\max}^{j,i}$ are the associated maximum distance in different stages. After replacing all $\rho_t^{i,j}$ in $\mathcal{B}_{\text{BE}}^{\mathbf{x}_t}$ of (13) by $\rho_{t,\min}^{i,j}$, we have $\mathcal{B}_{\text{BE},\min}^{\mathbf{x}_t}$, and similarly, we can get $\mathcal{B}_{\text{BE},\max}^{\mathbf{x}_t}$. Then, $\mathcal{B}_{\text{BE},\min}^{\mathbf{x}_t}\preceq\mathcal{B}_{\text{BE}}^{\mathbf{x}_t}\preceq\mathcal{B}_{\text{BE},\max}^{\mathbf{x}_t}$ is attained. By extracting $\rho_{t,\min}^{i,j}$ out of $\mathcal{B}_{\text{BE},\min}^{\mathbf{x}_t}$ (see (15) and (19)), we know $\mathcal{B}_{\text{BE},\min}^{\mathbf{x}_t}\propto(\rho_{t,\min}^{i,j})^4$. Similarly, $\mathcal{B}_{\text{BE},\max}^{\mathbf{x}_t}\propto(\rho_{t,\max}^{i,j})^4$ can be easily obtained. Combining with the above inequality, thus we have $\mathcal{B}_{\text{BE}}^{\mathbf{x}_t}\propto(\rho_{t,\max}^{i,j})^4$, as $\rho_{t,\min}^{i,j}\to\infty$. For $\mathcal{B}_{\text{BE}}^{\mathbf{r}_t^i}$, the result is got by obeying the above canon. Corollary 6 is proved.

Appendix G
Proof of Corollary 7

For the derivation of $\mathcal{B}_{\text{BE}}^{\mathbf{x}_t}$ in Appendix 1, the MLE-based FIM is directly dependent on the reference set Ω_t^j as per (80). Based on (97)–(101), all components of the construction $\mathcal{H}_{\text{BE}}^{\mathbf{x}_t}$ involve $\sum_{i\in\Omega_t^j}$. That is, when the uniform deployment holds, we get $\mathcal{H}_{\text{BE}}^{\mathbf{x}_t}\propto\Omega_t$ and thus $\mathcal{B}_{\text{BE}}^{\mathbf{x}_t}\propto\Omega_t^{-1}$. Similarly, we can get $\mathcal{B}_{\text{BE}}^{\mathbf{r}_t^i}\propto\Phi_t^{-1}$ and then (65) is proved. With the introduction of averaged observation information, the accuracy expressions in (13) and (38) can be re-derived as (64). Corollary 7 is proved.

Appendix H
Proof of Corollary 8

Based on (13) and (38), we can have $\mathcal{B}_{\text{BE}}^{\mathbf{x}_t}\propto(\mathcal{O}_{\text{BE}}^{\mathbf{x}_t})^{-1}$ and $\mathcal{B}_{\text{BE}}^{\mathbf{r}_t^i}\propto(\mathcal{R}_{\text{obs}}^{\mathbf{x}_t})^{-1}$. As per (14), (33), (40) and (50), we can get $\mathcal{O}_{\text{BE}}^{\mathbf{x}_t}\propto\mathbf{R}(\mathbf{r}_t)^{-1}, \mathbf{R}(\mathbf{r}_t)\propto(\mathbf{U}_t^i)^{-1}, \mathcal{R}_{\text{obs}}^{\mathbf{x}_t}\propto\mathrm{o}(\mathbf{x}_t)$ and $\mathrm{o}(\mathbf{x}_t)\propto\mathcal{H}_{\text{BE}}^{\mathbf{x}_t}$ easily. Thus, $\mathcal{B}_{\text{BE}}^{\mathbf{x}_t}\propto(\mathbf{U}_t^i)^{-1}$ and $\mathcal{B}_{\text{BE}}^{\mathbf{r}_t^i}\propto(\mathbf{U}_t^i)^{-1}$ can be got and then (68) is proved. For (67), it can be derived from (38). In terms of the first half of (66), we can further deduce that

$$\lim_{\mathbf{U}_t^i\to\mathbf{0},\forall i\in\Omega_\tau,\forall\tau=1:t}\mathcal{B}_{\text{BE}}^{\mathbf{x}_t} = \infty, \tag{106}$$

in which $\mathcal{B}_{\text{BE}}^{\mathbf{x}_t}\to\infty$ indicates that this VLP fails when all PD positions used for localization are completely incorrect. As for the latter half of (66), we have $\mathbf{R}(\mathbf{r}_t)\to\mathbf{0}$ as $\mathbf{U}_t^i\to\infty$, thus those elements after the minus sign in (15), (16), (17) and (18) will disappear. By (14), $\mathbf{F}(\mathbf{x}_t)\mathbf{K}(\mathbf{u})$ and $\mathbf{K}(\mathbf{u})^\top\mathbf{F}(\mathbf{x}_t)^\top$ can be separable common matrix factors, and then $\mathcal{L}_{\text{obs}}^{\mathbf{x}_t}$ is reworded as (69). Hence, we have confirmed (66). Corollary 8 is proved.

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